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Brand Names

Anacin Adult Low Strength, Aspergum, Aspir-Low, Aspirtab , Aspir-Trin , Bayer Advanced Aspirin, Bayer Aspirin, Bayer Aspirin Extra Strength, Bayer Aspirin Plus, Bayer Aspirin Regimen, Bayer Children's Aspirin, Bayer Extra Strength, Bayer Extra Strength Plus, Bayer Genuine Aspirin, Bayer Low Dose Aspirin Regimen, Bayer Womens Aspirin , BeneHealth Aspirin, Bufferin, Bufferin Extra Strength, Bufferin Low Dose, DURLAZA, Easprin , Ecotrin, Ecotrin Low Strength, Genacote, Halfprin, MiniPrin, NobleAid, St. Joseph Adult Low Strength, St. Joseph Aspirin, VAZALORE, Zero Order Release Aspirin, ZORprin

Indication Specific Dosing

General dosing information
Chewable, non-enteric coated tablets are preferred when rapid onset is required. If non-enteric coated tablets are unavailable, enteric-coated tablets may be chewed and swallowed for initial dosing.
In pediatric patients, dosage is typically rounded to the nearest 20.25 mg increment (e.g., one-fourth of an 81 mg tablet) based on measurable tablet size.

For the treatment of mild pain associated with arthralgia, dental pain, dysmenorrhea, headache, musculoskeletal pain (including backache), and/or the common cold

Oral dosage

Adults

325 or 650 mg PO every 4 hours as needed, or alternatively, 975 mg PO every 6 hours, as needed. Max: 3,900 mg/day. Discontinue use if pain gets worse or lasts more than 10 days.

Children and Adolescents 12 to 17 years

325 or 650 mg PO every 4 hours as needed, or alternatively, 975 mg PO every 6 hours, as needed. Max: 3,900 mg/day. Discontinue use if pain gets worse or lasts more than 10 days.

Rectal dosage

Adults

300 or 600 mg PR every 4 to 6 hours, as needed. Max: 3,600 mg/day. Discontinue use if pain gets worse or lasts more than 10 days.

Children and Adolescents 12 to 17 years

300 or 600 mg PR every 4 to 6 hours, as needed. Max: 3,600 mg/day. Discontinue use if pain gets worse or lasts more than 10 days.

For the treatment of acute ischemic stroke†

Oral dosage (immediate-release tablets or capsules)

Adults

160 to 325 mg PO once daily starting within 24 to 48 hours of stroke symptom onset. In patients with minor noncardioembolic ischemic stroke who did not receive a thrombolytic, use aspirin for first 21 days in combination with clopidogrel. Aspirin is not recommended as a substitute for acute stroke treatment in patients eligible for thrombolytic therapy or mechanical thrombectomy.

Infants, Children, and Adolescents

1 to 5 mg/kg/dose PO once daily. If dissection and cardioembolic causes are excluded, continue aspirin for a minimum of 2 years. Transition to clopidogrel, LMWH, or warfarin in those who have recurrent acute ischemic stroke (AIS) or transient ischemic attacks. For acute AIS due to non-Moyamoya vasculopathy, continue aspirin for 3 months; guide ongoing antithrombotic therapy with repeat cerebrovascular imaging.

Neonates

1 to 5 mg/kg/dose PO once daily. Aspirin is recommended for neonates with recurrent acute ischemic stroke.

Rectal dosage (suppository)

Adults

300 mg rectally once daily starting within 24 to 48 hours of stroke symptom onset. In patients with minor noncardioembolic ischemic stroke who did not receive a thrombolytic, use aspirin for first 21 days in combination with clopidogrel. Aspirin is not recommended as a substitute for acute stroke treatment in patients eligible for thrombolytic therapy or mechanical thrombectomy.

For secondary stroke prophylaxis after ischemic stroke or transient ischemic attack (TIA)

For stroke prophylaxis in persons with a history of noncardioembolic ischemic stroke or TIA

Oral dosage (immediate-release tablets or capsules)

Adults

50 to 325 mg PO once daily. Antiplatelet agents are recommended over oral anticoagulation.

For stroke prophylaxis in persons with a history of ischemic stroke or TIA and atrial fibrillation who are unable to take oral anticoagulants

Oral dosage (immediate-release tablets or capsules)

Adults

75 to 100 mg PO once daily in combination with clopidogrel.

For stroke prophylaxis and to reduce the risk of death in persons with a history of ischemic stroke or TIA

Oral dosage (extended-release capsules)

Adults

162.5 mg PO once daily.

For stroke prophylaxis in pediatric patients with a history of ischemic stroke or TIA†

Oral dosage (immediate-release tablets or capsules)

Infants, Children, and Adolescents

1 to 5 mg/kg/dose PO once daily for a minimum of 2 years. Transition to clopidogrel, LMWH, or warfarin in those who have recurrent acute ischemic stroke or transient ischemic attacks.

Neonates

1 to 5 mg/kg/dose PO once daily. Aspirin is recommended for neonates with recurrent acute ischemic stroke.

For stroke prophylaxis in pregnant or lactating persons in low-risk situations in which antiplatelet therapy is recommended

Oral dosage (immediate-release tablets or capsules)

Pregnant or Lactating Adults

50 to 150 mg PO once daily after the first trimester of pregnancy.

For stroke prophylaxis in persons with acute ischemic stroke (NIH stroke scale score 5 or less) or high risk TIA who are receiving ticagrelor

Oral dosage (immediate-release tablets or capsules)

Adults

300 to 325 mg PO loading dose, then 75 to 100 mg PO once daily plus ticagrelor for up to 30 days.

For the treatment of acute myocardial infarction, NSTEMI or unstable angina

Oral dosage

Adults

150 to 325 mg PO (non-enteric coated, chewable) once as soon as possible, then 75 to 162 mg PO once daily indefinitely. Maintenance doses up to 325 mg/day have been used in special circumstances. Aspirin is the established first-line therapy in patients with NSTE-ACS and reduces the risk of recurrent MI and death.

For the treatment of acute myocardial infarction, STEMI

Oral dosage

Adults

162 to 325 mg PO (non-enteric coated, chewable) once as soon as possible, then 75 to 100 mg PO once daily. Guidelines recommend aspirin with a P2Y12 inhibitor for at least 12 months in individuals who are not at high bleeding risk. Discontinuation of aspirin before 12 months may be appropriate in select individuals. Higher maintenance doses of up to 325 mg/day have been used but may be associated with a higher risk of bleeding.

For reduction of cardiovascular mortality (e.g., myocardial infarction prophylaxis, stroke prophylaxis, thrombosis prophylaxis, arterial thromboembolism prophylaxis)

For primary prevention of atherosclerotic cardiovascular disease (ASCVD)

Oral dosage

Adults

75 to 162 mg PO once daily. Consider in persons 40 to 70 years who are high risk ASCVD but not for bleeding.

For secondary prevention of atherosclerotic cardiovascular disease (ASCVD)

Oral dosage

Adults

75 to 162 mg PO once daily indefinitely for all persons with CAD unless contraindicated. In persons treated with dual antiplatelet therapy (DAPT) or triple therapy (aspirin, P2Y12 inhibitor, and oral anticoagulant), 81 mg PO once daily is the recommended dose.

For secondary prevention of atherosclerotic cardiovascular disease (ASCVD) in persons undergoing coronary artery bypass graft (CABG) surgery

Oral dosage

Adults

75 to 325 mg PO preoperatively and within 6 hours after CABG; continue once

daily dosing indefinitely.

For primary prevention of cardiovascular events in persons with asymptomatic peripheral artery disease or carotid artery stenosis

Oral dosage

Adults

75 to 325 mg PO once daily.

For secondary prevention of cardiovascular events in persons with symptomatic peripheral artery disease or carotid artery stenosis (including recent carotic endarterectomy)

Oral dosage

Adults

75 to 325 mg PO once daily.

For general antiplatelet therapy† and thromboprophylaxis† in pediatric populations

Oral dosage

Infants, Children, and Adolescents

1 to 5 mg/kg/dose (Max: 81 to 325 mg/dose) PO once daily.

Neonates

1 to 5 mg/kg/dose PO once daily.

For postoperative thromboprophylaxis† in pediatric populations undergoing Blalock-Taussig (BT) shunt placement†, Glenn procedure†, Norwood procedure†, Sano procedure†, or Fontan procedure†

Oral dosage

Infants and Children

1 to 5 mg/kg/dose (Max: 81 to 325 mg/day) PO once daily. Higher doses (up to 10 mg/kg/day) have been reported. There was no significant difference in thrombosis rate at 2 years in children receiving warfarin or aspirin (24% vs. 14%,

p = 0.45) in a multicenter, randomized control trial of 111 children after Fontan surgery. Although not statistically significant, the incidence of thrombosis was 9% in children receiving aspirin (n = 34) compared to 2% in children receiving rivaroxaban (n = 64) for thromboprophylaxis post-Fontan procedure in a randomized, multicenter, open-label study.

Neonates

1 to 5 mg/kg/dose PO once daily. Higher doses (up to 15 mg/kg/day) have been reported. A flat dose of 40 mg/day PO was described in a retrospective review evaluating thrombosis after modified BT shunt placement in 207 children, 162 (78%) which were neonates, with a mean weight of 3.1 +/- 0.8 kg.

For postoperative thromboprophylaxis† in pediatric populations undergoing atrial or ventricular septal defect repair†

Oral dosage

Infants, Children, and Adolescents

1 to 5 mg/kg/dose (Max: 81 to 325 mg/dose) PO once daily; begin 1 to several days before device implantation and continue for at least 6 months. Higher doses (up to 10 mg/kg/day) have been reported.

For thromboprophylaxis† in pediatric populations with ventricular assist devices (VADs)†

Oral dosage

Infants, Children, and Adolescents

1 to 5 mg/kg/dose PO once daily; begin within 72 hours of VAD placement. Use in combination with heparin (begun 8 to 48 hours after implantation) and with or without dipyridamole.

For thromboprophylaxis in persons with valvular heart disease† and prosthetic heart valves†

Oral dosage

Adults

75 to 100 mg PO once daily indefinitely.

For primary thromboprophylaxis in persons with systemic lupus erythematosus† without antiphospholipid syndrome but with high-risk antiphospholipid antibody profile

Oral dosage

Adults

75 to 100 mg PO once daily. A high-risk antiphospholipid antibody profile is defined as lupus anticoagulant positivity, or double (any combination of lupus anticoagulant, anticardiolipin antibodies, or anti-beta2glycoprotein I antibodies), or triple antiphospholipid antibodies (all 3).

For the secondary prevention of cardiovascular events in patients with intermittent claudication†

Oral dosage

Adults

75 to 325 mg PO once daily.

For the prevention of ischemic complications in patients undergoing percutaneous coronary intervention (PCI)†

Oral dosage

Adults

300 to 325 mg PO (non-enteric coated) at least 2 hours (preferably 24 hours) before PCI; for patients already on daily aspirin, 75 to 325 mg PO before PCI. Continue 75 to 81 mg PO once daily indefinitely. Lower loading doses (150 to 162 mg) and higher maintenance doses (up to 325 mg/day) may be used.

For colorectal cancer prophylaxis†

Oral dosage

Adults 50 to 69 years

81 mg PO once daily for patients who have a 10% or more 10-year CVD risk, are not at increased risk for bleeding, have a life expectancy of at least 10 years, and are willing to take low-dose aspirin daily for at least 10 years.

For the treatment of fever

Oral dosage

Adults

325 or 650 mg PO every 4 hours as needed, or alternatively, 975 mg PO every 6 hours, as needed. Max: 3,900 mg/day. Discontinue use if fever gets worse or lasts more than 3 days.

Children and Adolescents 12 to 17 years

325 or 650 mg PO every 4 hours as needed, or alternatively, 975 mg PO every 6 hours, as needed. Max: 3,900 mg/day. Discontinue use if fever gets worse or lasts more than 3 days.

Rectal dosage

Adults

300 or 600 mg PR every 4 to 6 hours, as needed. Max: 3,600 mg/day. Discontinue use if fever gets worse or lasts more than 3 days.

Children and Adolescents 12 to 17 years

300 or 600 mg PR every 4 to 6 hours, as needed. Max: 3,600 mg/day. Discontinue use if fever gets worse or lasts more than 3 days.

For the treatment of Kawasaki disease†

Oral dosage (high-dose)

Infants, Children, and Adolescents

80 to 100 mg/kg/day PO divided every 6 hours during the acute phase, then 3 to 5 mg/kg/dose (Max: 325 mg/dose) PO once daily for 4 to 8 weeks after the onset of illness. Duration of high-dose aspirin varies in clinical practice; while many clinicians reduce the aspirin dose after the individual is afebrile for 24 to 72 hours, others continue high-dose aspirin until Day 14 of illness and at least 48 to 72 hours after cessation of fever. There is also debate over the optimal dose of aspirin in the acute phase of treatment. Data show that high-dose aspirin in the acute phase is likely not associated with improved coronary artery outcomes. Additionally, some data suggest low-dose aspirin is not inferior to high-dose

aspirin in reducing the risk of coronary artery aneurysms when given concomitantly with IVIG during the acute phase.

Oral dosage (moderate-dose)

Infants, Children, and Adolescents

30 to 50 mg/kg/day PO divided every 6 hours during the acute phase, then 3 to 5 mg/kg/dose (Max: 325 mg/dose) PO once daily for at least 4 to 8 weeks after the onset of illness. Duration of moderate-dose aspirin varies in clinical practice; many clinicians reduce the aspirin dose after the individual is afebrile for at least 48 to 72 hours. There is also debate over the optimal dose of aspirin in the acute phase of treatment. Data show that moderate-dose aspirin in the acute phase is likely not associated with improved coronary artery outcomes.

Oral dosage (low-dose)

Infants, Children, and Adolescents

3 to 10 mg/kg/dose (Max: 325 mg/dose) PO once daily for at least 4 to 8 weeks after the onset of illness. For those who develop coronary abnormalities, low-dose therapy may continue indefinitely. Some data suggest low-dose aspirin is not inferior to high-dose aspirin in reducing the risk of coronary artery aneurysms when given concomitantly with IVIG during the acute phase.

For the treatment of acute or recurrent pericarditis†

For the treatment of acute pericarditis†

Oral dosage

Adults

750 to 1,000 mg PO every 8 hours for 1 to 2 weeks, then decrease dose by 250 to 500 mg/day every 1 to 2 weeks in combination with colchicine.

For the treatment of recurrent pericarditis†

Oral dosage

Adults

500 to 1,000 mg PO every 6 to 8 hours for at least 2 to 4 weeks, then decrease dose by 250 to 500 mg/day every 1 to 2 weeks in combination with colchicine. Dose range: 1.5 to 4 g/day.

For migraine prophylaxis†

Oral dosage

Adults

100 or 325 mg PO every other day, or alternately, 300 mg PO once daily. Clinical practice guidelines classify aspirin as having inadequate or conflicting data to support or refute use for migraine prophylaxis.

For preeclampsia prophylaxis†

For preeclampsia prophylaxis† in high-risk pregnant patients

Oral dosage

Adults

81 mg PO once daily starting at 12 weeks gestation. Consider 162 mg PO once daily with pre-existing diabetes mellitus.

Adolescents

81 mg PO once daily starting at 12 weeks gestation. Consider 162 mg PO once daily with pre-existing diabetes mellitus.

For preeclampsia prophylaxis† in pregnant patients with low to moderate risk

Oral dosage

Adults

65 to 100 mg PO once daily initiated at various times during pregnancy (e.g., 12 weeks) and continued until 34 weeks or delivery has been used.

Adolescents

65 to 100 mg PO once daily initiated at various times during pregnancy (e.g., 12 weeks) and continued until 34 weeks or delivery has been used.

For the acute treatment of migraine†

Oral dosage (immediate-release tablets or capsules)

Adults

500 mg PO as a single dose. Guidelines classify aspirin as having established efficacy for the treatment of acute migraine.

For the management of multisystem inflammatory syndrome in children (MIS-C) post SARS-CoV-2 exposure†

Oral dosage

Infants, Children, and Adolescents

3 to 5 mg/kg/dose (Max: 81 mg) PO once daily for all patients without risk factors for bleeding. Continuation is recommended until platelet count is normalized and normal coronary arteries are confirmed at least 4 weeks after diagnosis. Avoid use in patients with active bleeding, significant bleeding risk, and/or a platelet count of 80,000/microliter or less. Patients with coronary artery aneurysms and a maximal z-score of 2.5 to 10 should be treated with low dose aspirin, whereas patients with a z-score of 10 or more should be treated with low dose aspirin and therapeutic anticoagulation with enoxaparin for at least 2 weeks before transitioning to warfarin. Patients with an ejection fraction (EF) less than 35% should receive low dose aspirin and therapeutic anticoagulation until EF exceeds 35%. Patients with documented thrombosis should receive low dose aspirin and therapeutic anticoagulation for 3 months, pending thrombosis resolution.

For the treatment of rheumatic arthritis†

Oral dosage (immediate-release tablets or capsules)

Adults

50 to 60 mg/kg/day PO in 4 to 5 divided doses, initially. May increase the dose up to 80 to 100 mg/kg/day (Usual dose: 4 to 8 g/day) based on serum aspirin concentrations (20 to 30 mg/dL), clinical response, and tolerability. If there is an incomplete response within 2 weeks, may increase the dose to 125 mg/kg/day. Treat for 1 month or until inflammation has subsided.

Infants, Children, and Adolescents

50 to 60 mg/kg/day PO in 4 to 5 divided doses, initially. May increase the dose up to 80 to 100 mg/kg/day based on serum aspirin concentrations (20 to 30 mg/dL), clinical response, and tolerability. If there is an incomplete response within 2

weeks, may increase the dose to 125 mg/kg/day. Treat for 1 month or until inflammation has subsided.

Contraindications And Precaution

Drug Interactions

The coadministration of certain medications may lead to harm and require avoidance or therapy modification; review all drug interactions prior to concomitant use of other medications.

Hypersensitivity

This medication is contraindicated in patients with a history of hypersensitivity to it or any of its components. Aspirin is also contraindicated in individuals with a history of hypersensitivity to nonsteroidal anti-inflammatory drugs.

Treatment-Related Restrictions of Use

Use aspirin with caution in individuals with cardiovascular disease. Observational data from a national registry demonstrated that individuals treated with NSAIDs in the post-myocardial infarction period were at increased risk of reinfarction, cardiovascular (CV)-related death, and all-cause mortality beginning the first week of treatment. Individuals with known cardiovascular disease or risk factors had a higher absolute incidence of excess serious CV thrombotic events, due to their increased baseline rate. Some observational studies found that this increased risk of serious CV thrombotic events began as early as the first weeks of treatment with an NSAID; the increase in CV thrombotic risk has been observed most consistently at higher doses. Current evidence is insufficient to determine if the risk of an event is higher or lower for any particular NSAID compared to other NSAIDs. There is no consistent evidence that concomitant use of aspirin mitigates the increased risk for CV thrombotic events. To minimize the potential risk for an adverse CV event in NSAID-treated individuals, use the lowest effective dose for the shortest duration possible; remain alert for the development of such events, throughout the entire treatment course, even in the absence of previous CV symptoms. NSAIDs may worsen pre-existing hypertension, which may contribute to the increased incidence of CV events. Monitor blood pressure during NSAID treatment initiation and throughout therapy.

aspirin exacerbated respiratory disease, asthma

Aspirin is contraindicated in individuals with aspirin exacerbated respiratory disease, which may include chronic rhinosinusitis complicated by nasal polyps, severe and potentially fatal acute bronchospasm, and/or intolerance to aspirin and other NSAIDs.

When aspirin is used in individuals with pre-existing asthma without known aspirin sensitivity, monitor for changes in the signs and symptoms of asthma.

bleeding disorder, ethanol ingestion, neonates, peptic ulcer disease

Avoid aspirin in individuals with active peptic ulcer disease due to the risk of gastric ulceration and bleeding. Aspirin increases bleeding risk; risk factors for bleeding include history of bleeding disorder, ethanol ingestion, age 60 years and older, and the use of other drugs that increase the risk of bleeding (e.g., anticoagulants, antiplatelet agents, and chronic NSAID therapy). Neonates have a slower clearance of aspirin and therefore are at higher risk for bleeding.

influenza, varicella

Do not use aspirin in pediatric populations who have or are recovering from varicella (chicken pox), influenza, or influenza-like symptoms due to the risk of Reye syndrome.

hepatic failure

Avoid aspirin in individuals with severe hepatic insufficiency or hepatic failure.

renal disease

Avoid aspirin in individuals with severe renal failure (i.e., GFR less than 10 mL/minute). However, guidelines recommend aspirin for primary and secondary prevention of atherosclerotic events with careful monitoring for bleeding complications in individuals undergoing dialysis. Aspirin use decreases GFR and renal blood flow, especially in individuals with pre-existing renal disease. The regular use of aspirin is associated with a dose-dependent increased risk of chronic renal failure.

geriatric

Aspirin increases bleeding risk; risk factors for bleeding include age 60 years and older. According to the Beers Criteria, aspirin is considered a potentially inappropriate medication for use in geriatric adults. In general, avoid the use of aspirin as primary prevention of cardiovascular (CV) disease in older adults. Several studies suggest a lack of net benefit when used for primary prevention of CV disease in older adults and the risk of gastrointestinal (GI) bleeding increases markedly in older age. Consider the deprescribing of aspirin in older adults currently taking it for primary CV prevention. Aspirin is generally indicated for secondary CV prevention in older adults with established CV disease. Aspirin may cause new or worsening gastric or duodenal ulcers, and there is an increased risk of GI bleeding and peptic ulcer disease in high-risk groups

including those more than 75 years of age, or those taking oral or parenteral corticosteroids, anticoagulants, or antiplatelet medications. The risk of ulcers, gross bleeding, or perforation is cumulative with continued use. The Beers panel also recommends avoiding chronic use of aspirin doses more than 325 mg/day in high-risk individuals, including those with a history of gastric or duodenal ulcers, unless other alternatives are not effective and the individual can take a gastroprotective agent. The use of a gastroprotective agent, like a proton-pump inhibitor or misoprostol, reduces but does not eliminate, GI risks.

labor, obstetric delivery, pregnancy

Daily low-dose aspirin use is considered safe during pregnancy with a low likelihood of serious complications. Guidelines recommend prophylactic low-dose aspirin after 12 weeks of gestation in people at high risk for preeclampsia. The use of NSAIDs, including aspirin for other indications and at doses other than those recommended for preeclampsia prophylaxis, is generally not recommended during the second half of pregnancy. Avoid the use of NSAIDs during pregnancy starting at 30 weeks gestation due to the risk of premature closure of the fetal ductus arteriosus. When use is necessary between 20 and 30 weeks gestation, limit use to the lowest effective dose and shortest duration possible. Consider ultrasound monitoring of amniotic fluid if treatment exceeds 48 hours. Discontinue the NSAID if oligohydramnios develops, and follow up according to clinical practice. Use of NSAIDs around 20 weeks gestation and later may also cause fetal renal dysfunction, leading to oligohydramnios, and in some cases, neonatal renal impairment. These adverse outcomes are seen, on average, after days to weeks of treatment, although oligohydramnios has been infrequently reported as soon as 48 hours after NSAID initiation. Oligohydramnios is often, but not always, reversible with treatment discontinuation. Complications of prolonged oligohydramnios may include limb contractures and delayed lung maturation. In some cases, invasive neonatal procedures such as exchange transfusion or dialysis have been required. Salicylates have also been associated with alterations in maternal and neonatal hemostasis mechanisms, decreased birth weight, and perinatal mortality. Avoid the use of aspirin for indications and at doses other than those recommended for preeclampsia prophylaxis 1 week prior to and during labor and obstetric delivery due to the potential for excessive blood loss at delivery. Prolonged gestation and labor due to prostaglandin inhibition have been reported.

breast-feeding

Use aspirin with caution during breast-feeding. Daily low-dose (e.g., 75 to 162 mg/day) aspirin therapy for vascular indications may be continued during breast-feeding. Aspirin is not present in human milk after daily low-dose aspirin (75 to 325 mg/day) use and

salicylate concentrations are low. Long term use of higher doses are not recommended as salicylates are excreted into human milk, with higher doses resulting in disproportionately higher milk levels which could cause adverse effects in the breastfed child. Mean peak breast milk concentrations of salicylate in 6 lactating individuals after aspirin doses of 500, 1,000, and 1,500 mg were 5.8, 15.8, and 38.8 mg/L, respectively. Salicylate concentrations were detectable in breast milk within 1 hour of dosing and reached maximum concentration within 2 to 6 hours. There has been a case report linking long-term, high dose aspirin use by a lactating person to metabolic acidosis in a 16-day-old neonate. If aspirin is used during lactation, monitor the breastfed child for bruising and bleeding. Alternative analgesics and antipyretics considered to be usually compatible with breast-feeding include acetaminophen and ibuprofen.

people who may become pregnant, reproductive risk

Counsel people who may become pregnant about the reproductive risk associated with aspirin. NSAIDs may delay or prevent prostaglandin-mediated rupture of ovarian follicles, which has been associated with reversible infertility. Small studies with NSAIDs demonstrated a reversible delay in ovulation. Consider withdrawal of NSAIDs in people who may become pregnant who have difficulties conceiving or who are undergoing infertility evaluation.

adolescents, children, infants

The Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list includes aspirin as a potentially inappropriate medication for use in infants, children, and adolescents with suspicion of a viral illness and use should be avoided. If avoidance is not possible, use aspirin with caution and appropriate monitoring.

Pregnancy And Lactation

Use aspirin with caution during breast-feeding. Daily low-dose (e.g., 75 to 162 mg/day) aspirin therapy for vascular indications may be continued during breast-feeding. Aspirin is not present in human milk after daily low-dose aspirin (75 to 325 mg/day) use and salicylate concentrations are low. Long term use of higher doses are not recommended as salicylates are excreted into human milk, with higher doses resulting in disproportionately higher milk levels which could cause adverse effects in the breastfed child. Mean peak breast milk concentrations of salicylate in 6 lactating individuals after aspirin doses of 500, 1,000, and 1,500 mg were 5.8, 15.8, and 38.8 mg/L, respectively. Salicylate concentrations were detectable in breast milk within 1 hour of dosing and reached maximum concentration within 2 to 6 hours. There has been a case report

linking long-term, high dose aspirin use by a lactating person to metabolic acidosis in a 16-day-old neonate. If aspirin is used during lactation, monitor the breastfed child for bruising and bleeding. Alternative analgesics and antipyretics considered to be usually compatible with breast-feeding include acetaminophen and ibuprofen.

Interactions

Abrocitinib: (Contraindicated) Concurrent use with daily aspirin doses higher than 81 mg is contraindicated during the first 3 months of abrocitinib therapy due to an increased risk of bleeding with thrombocytopenia.
Acarbose: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood sugar. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Acebutolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Acetaminophen; Aspirin, ASA; Caffeine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Acetaminophen; Caffeine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Acetaminophen; Caffeine; Dihydrocodeine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Acetaminophen; Caffeine; Pyrilamine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Acetaminophen; Ibuprofen: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of

ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6 days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen 400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.
acetaZOLAMIDE: (Major) Avoid the coadministration of high-dose salicylates and carbonic anhydrase inhibitors whenever possible. There were reports of anorexia, tachypnea, lethargy, metabolic acidosis, coma, and death with high-dose aspirin and acetazolamide. Two mechanisms could cause increased acetazolamide concentrations, resulting in CNS depression and metabolic acidosis: first, competition with aspirin for renal tubular secretion and, second, displacement by salicylates from plasma protein binding sites. Additionally, carbonic anhydrase inhibitors alkalinize urine and increase the excretion of normal doses of salicylates; decreased plasma salicylate concentrations may or may not be clinically significant.
Acetohydroxamic Acid: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Acidifying Agents: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Ado-Trastuzumab emtansine: (Moderate) Use caution if coadministration of aspirin with ado-trastuzumab emtansine is necessary due to reports of severe and sometimes fatal hemorrhage, including intracranial bleeding, with ado-trastuzumab emtansine therapy. Consider additional monitoring when concomitant use is medically necessary. While some patients who experienced bleeding during ado-trastuzumab therapy were also

receiving anticoagulation therapy, others had no known additional risk factors.
Albuterol; Budesonide: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Alendronate: (Minor) Monitor for gastrointestinal adverse events during concurrent use of alendronate and aspirin. Both medications have been associated with gastrointestinal irritation although data suggest concomitant use introduces little additional risk for adverse effects for most patients.
Alendronate; Cholecalciferol: (Minor) Monitor for gastrointestinal adverse events during concurrent use of alendronate and aspirin. Both medications have been associated with gastrointestinal irritation although data suggest concomitant use introduces little additional risk for adverse effects for most patients.
Aliskiren; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Alkalinizing Agents: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Alogliptin: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood glucose concentrations. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Alogliptin; metFORMIN: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood glucose concentrations. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Alogliptin; Pioglitazone: (Moderate) Monitor blood glucose during concomitant

thiazolidinedione and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood glucose concentrations. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Alpha-glucosidase Inhibitors: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood sugar. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Alteplase: (Moderate) Monitor for bleeding if the concomitant use of thrombolytics and aspirin is necessary. For patients who received a thrombolytic for acute ischemic stroke, avoid IV aspirin in the first 90 minutes following thrombolytic administration and consider delaying the initiation of aspirin therapy for at least 24 hours following thrombolytic administration. Both medications interfere with hemostasis and concomitant use increases the risk for bleeding.
Amikacin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
aMILoride: (Moderate) Salicylates can increase the risk of renal insufficiency in patients receiving diuretics, secondary to effects on renal blood flow. Salicylates inhibit renal prostaglandin production, which causes salt and water retention and decreased renal blood flow. Coadministration may cause hyperkalemia.
aMILoride; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention. (Moderate) Salicylates can increase the risk of renal insufficiency in patients receiving diuretics, secondary to effects on renal blood flow. Salicylates inhibit renal prostaglandin production, which causes salt and water retention and decreased renal blood flow. Coadministration may cause hyperkalemia.
Aminoglycosides: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
Aminolevulinic Acid: (Minor) Preclinical data suggest that agents that affect platelet

function and inhibit prostaglandin synthesis could decrease the efficacy of photosensitizing agents used during photodynamic therapy.
Amivantamab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
amLODIPine; Benazepril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
amLODIPine; Celecoxib: (Major) Concomitant use of analgesic doses of aspirin and celecoxib is generally not recommended due to the increased risk of bleeding. Concurrent use of analgesic doses of aspirin with NSAIDs does not produce a greater therapeutic effect compared to the use of NSAIDs alone. Celecoxib (200 to 400 mg/day) did not interfere with the cardioprotective antiplatelet effect of aspirin (100 to 325 mg) in 2 studies in healthy volunteers and in patients with osteoarthritis and established heart disease. Celecoxib is not a substitute for low dose aspirin for cardiovascular protection.
amLODIPine; Valsartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Amobarbital: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Amoxicillin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Amoxicillin; Clarithromycin; Omeprazole: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination

should be used with caution and patients monitored for increased side effects.
Amoxicillin; Clavulanic Acid: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Amphotericin B lipid complex (ABLC): (Minor) Concurrent use of amphotericin B and other potentially nephrotoxic medications, like the salicylates, may enhance the potential for drug-induced renal toxicity.
Amphotericin B liposomal (LAmB): (Minor) Concurrent use of amphotericin B and other potentially nephrotoxic medications, like the salicylates, may enhance the potential for drug-induced renal toxicity.
Amphotericin B: (Minor) Concurrent use of amphotericin B and other potentially nephrotoxic medications, like the salicylates, may enhance the potential for drug-induced renal toxicity.
Ampicillin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Ampicillin; Sulbactam: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Anagrelide: (Moderate) Use caution with the coadministration of aspirin and anagrelide. The coadministration of single or repeated doses of anagrelide and aspirin resulted in greater ex vivo anti-platelet aggregation effects than administration of aspirin alone. In an observational study, the concomitant use of anagrelide and aspirin increased the rate of major hemorrhagic events compared to patients receiving other cytoreductive therapy. Assess the risks and benefits of concomitant aspirin and anagrelide use, particularly in patients at high risk for hemorrhage. Monitor for bleeding during concomitant therapy.
Angiotensin-converting enzyme inhibitors: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of

the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Antithrombin III: (Moderate) Large doses of salicylates (more than 3 to 4 g/day) can cause hypoprothrombinemia, an additional risk factor for bleeding. Patients taking large doses of salicylates and antithrombin III should be monitored closely for bleeding.
Apixaban: (Major) Large doses of salicylates (3 to 4 g/day or more) can cause hypoprothrombinemia, an additional risk factor for bleeding. Patients taking large doses of salicylates and apixaban should be monitored closely for bleeding.
Argatroban: (Moderate) An additive risk of bleeding may be seen in patients receiving salicylates (e.g., aspirin) in combination with thrombin inhibitors. In addition, large doses of salicylates (3 to 4 g/day or more) can cause hypoprothrombinemia, an additional risk factor for bleeding. Nonsteroidal antiinflammatory drugs (NSAIDs) may also increase bleeding risk when given with argatroban because of their potential to cause GI bleeding or inhibit platelet aggregation.
Ascorbic Acid, Vitamin C: (Minor) Agents that acidify the urine should be avoided in patients receiving high-dose salicylates. Urinary pH changes can decrease salicylate excretion. However, if the urine is acidic prior to administration of an acidifying agent, the increase in salicylic acid concentrations should be minimal.
Aspirin, ASA; Butalbital; Caffeine: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur. (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Aspirin, ASA; Caffeine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Aspirin, ASA; Caffeine; Orphenadrine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Aspirin, ASA; Citric Acid; Sodium Bicarbonate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Aspirin, ASA; Dipyridamole: (Moderate) Monitor for bleeding during concomitant use of aspirin and dipyridamole; both medications interfere with platelet aggregation and concomitant use increases the risk for bleeding. For patients undergoing a dipyridamole nuclear stress test and receiving combination aspirin; dipyridamole, hold aspirin; dipyridamole for 48 hours prior to the stress test; concomitant use may increase the risk for cardiovascular side effects and impair the sensitivity of the test.
Atenolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Atenolol; Chlorthalidone: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.

(Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Atezolizumab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Azelastine; Fluticasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Azilsartan; Chlorthalidone: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Bacitracin: (Minor) Additive nephrotoxicity may occur with concurrent use of systemic bacitracin and other nephrotoxic agents, including salicylates. Topical administration of any preparation containing bacitracin, especially when applied to large surface areas, also should not be given with other drugs that have a nephrotoxic potential.
Barbiturates: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Beclomethasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Benazepril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart

disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Benazepril; hydroCHLOROthiazide, HCTZ: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Beta-blockers: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Betamethasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Betaxolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Bismuth Subsalicylate: (Moderate) Monitor for salicylate-related adverse effects, including salicylate toxicity, if concomitant use of aspirin and bismuth subsalicylate is necessary. Adverse reactions, such as bleeding, renal impairment, and tinnitus, may occur.
Bismuth Subsalicylate; metroNIDAZOLE; Tetracycline: (Moderate) Monitor for salicylate-related adverse effects, including salicylate toxicity, if concomitant use of aspirin and bismuth subsalicylate is necessary. Adverse reactions, such as bleeding, renal impairment, and tinnitus, may occur.
Bisoprolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins

and thus, salt and water retention and decreased renal blood flow.
Bisoprolol; hydroCHLOROthiazide, HCTZ: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Bivalirudin: (Moderate) An additive risk of bleeding may be seen in patients receiving salicylates (e.g., aspirin) in combination with thrombin inhibitors. In addition, large doses of salicylates (3 to 4 g/day or more) can cause hypoprothrombinemia, an additional risk factor for bleeding. Nonsteroidal antiinflammatory drugs (NSAIDs) may also increase bleeding risk when given with argatroban because of their potential to cause GI bleeding or inhibit platelet aggregation.
Brimonidine; Timolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Bromocriptine: (Minor) Bromocriptine is highly bound (more than 90%) to serum proteins. Therefore, it may increase the unbound fraction of other highly protein-bound medications (e.g., aspirin and other salicylates), which may alter their effectiveness and risk for side effects.
Budesonide: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Budesonide; Formoterol: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Budesonide; Glycopyrrolate; Formoterol: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids

enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Bumetanide: (Moderate) Salicylates may decrease the diuretic, natriuretic, and antihypertensive actions of diuretics, possibly through inhibition of renal prostaglandin synthesis. Patients receiving loop diuretics and salicylates should be monitored for changes in the effectiveness of their diuretic therapy.
BUPivacaine; Meloxicam: (Major) Concomitant use of low dose aspirin or analgesic doses of aspirin and meloxicam is generally not recommended due to the increased risk of bleeding and renal impairment. Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Meloxicam is not a substitute for low dose aspirin for cardiovascular protection.
busPIRone: (Minor) In vitro studies showed that therapeutic levels of aspirin, ASA increased the plasma concentrations of free buspirone by 23% through plasma protein binding displacement. In vivo interaction studies with these drugs have not been performed.
Butalbital; Acetaminophen: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Butalbital; Acetaminophen; Caffeine: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur. (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Butalbital; Acetaminophen; Caffeine; Codeine: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur. (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Butalbital; Aspirin; Caffeine; Codeine: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur. (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Caffeine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Caffeine; Sodium Benzoate: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Canagliflozin: (Moderate) Monitor blood glucose during concomitant canagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Canagliflozin; metFORMIN: (Moderate) Monitor blood glucose during concomitant

canagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Candesartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Caplacizumab: (Major) Avoid concomitant use of caplacizumab and aspirin when possible. Assess and monitor closely for bleeding if use together is necessary. Interrupt use of caplacizumab if clinically significant bleeding occurs.
Capreomycin: (Major) Since capreomycin is eliminated by the kidney, coadministration with other potentially nephrotoxic drugs, including salicylates, may increase serum concentrations of either drug. Theoretically, the chronic coadministration of these drugs may increase the risk of developing nephrotoxicity, even in patients who have normal renal function. Monitor patients for changes in renal function if these drugs are coadministered.
Captopril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Captopril; hydroCHLOROthiazide, HCTZ: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may

diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Carteolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Carvedilol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Cefixime: (Minor) In vitro, salicylates have displaced cefixime from its protein-binding sites, resulting in a 50% increase in free cefixime levels. The clinical significance of this effect is unclear at this time.
cefoTEtan: (Minor) Cefotetan has been associated with hypoprothrombinemia and may cause additive effects when given concurrently with salicylates.
Celecoxib: (Major) Concomitant use of analgesic doses of aspirin and celecoxib is generally not recommended due to the increased risk of bleeding. Concurrent use of analgesic doses of aspirin with NSAIDs does not produce a greater therapeutic effect compared to the use of NSAIDs alone. Celecoxib (200 to 400 mg/day) did not interfere with the cardioprotective antiplatelet effect of aspirin (100 to 325 mg) in 2 studies in healthy volunteers and in patients with osteoarthritis and established heart disease. Celecoxib is not a substitute for low dose aspirin for cardiovascular protection.
Celecoxib; Tramadol: (Major) Concomitant use of analgesic doses of aspirin and celecoxib is generally not recommended due to the increased risk of bleeding. Concurrent use of analgesic doses of aspirin with NSAIDs does not produce a greater therapeutic effect compared to the use of NSAIDs alone. Celecoxib (200 to 400 mg/day) did not interfere with the cardioprotective antiplatelet effect of aspirin (100 to 325 mg) in 2 studies in healthy volunteers and in patients with osteoarthritis and established heart disease. Celecoxib is not a substitute for low dose aspirin for cardiovascular protection.
Chlorothiazide: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Chlorpheniramine; Ibuprofen; Pseudoephedrine: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or

non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6 days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen 400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.

Chlorthalidone: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.

Ciclesonide: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of

renal clearance.
Cidofovir: (Contraindicated) The concomitant administration of cidofovir and NSAIDs, such as aspirin, is contraindicated due to the potential for increased nephrotoxicity. Aspirin should be discontinued 7 days prior to beginning cidofovir.
Cilostazol: (Moderate) Use caution with the coadministration of aspirin and cilostazol. Although the short-term (<= 4 days) coadministration of aspirin and cilostazol increased the inhibition of ADP-induced platelet aggregation by 22% to 37% compared to aspirin or cilostazol use alone, no clinically significant effect on PT, aPTT, or bleeding time was observed compared to aspirin alone. In clinical trials, there was no apparent increase in hemorrhagic adverse effects in patients taking cilostazol and aspirin compared to aspirin alone. The effects of long-term coadministration are unknown. Monitor for bleeding during concomitant therapy.
Citalopram: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Citric Acid; Potassium Citrate; Sodium Citrate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid. (Moderate) Urinary alkalinizing agents, like potassium citrate, increase the excretion of salicylates by increasing renal clearance.
clomiPRAMINE: (Moderate) Clomipramine may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. This may increase the risk for an upper GI bleed.
Clopidogrel: (Moderate) Monitor for bleeding if aspirin and clopidogrel are used together as concomitant has an additive effect on platelet function.
Colistimethate, Colistin, Polymyxin E: (Major) Theoretically, the chronic coadministration of these drugs may increase the risk of developing nephrotoxicity, even in patients who have normal renal function. Monitor patients for changes in renal function if these drugs are coadministered. Since colistimethate sodium is eliminated by the kidney, coadministration with other potentially nephrotoxic drugs, including salicylates, may increase serum concentrations of either drug.
Colistin: (Major) Theoretically, the chronic coadministration of these drugs may increase the risk of developing nephrotoxicity, even in patients who have normal renal function. Monitor patients for changes in renal function if these drugs are coadministered. Since colistimethate sodium is eliminated by the kidney, coadministration with other potentially nephrotoxic drugs, including salicylates, may increase serum concentrations of either drug.
Collagenase: (Moderate) Cautious use of injectable collagenase by patients taking more than 150 mg/day of aspirin is advised. The efficacy and safety of administering injectable

collagenase to a patient taking more than 150 mg/day of aspirin within 7 days before the injection are unknown. Receipt of injectable collagenase may cause an ecchymosis or bleeding at the injection site.
Corticosteroids: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Cortisone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
cycloSPORINE: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like cyclosporine may lead to additive nephrotoxicity.
Dabigatran: (Major) Educate patients about the signs of increased bleeding and the need to report these signs to a healthcare provider immediately if coadministration of dabigatran and aspirin or another salicylate is necessary. Dabigatran can cause significant and, sometimes, fatal bleeding. This risk may be increased by concurrent use of chronic salicylate therapy.
Dalteparin: (Moderate) An additive risk of bleeding may be seen in patients receiving a low molecular weight heparin in combination with other agents known to increase the risk of bleeding such as salicylates. Monitor clinical and laboratory response closely during concurrent use.
Dapagliflozin: (Moderate) Monitor blood glucose during concomitant dapagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Dapagliflozin; metFORMIN: (Moderate) Monitor blood glucose during concomitant dapagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Dapagliflozin; sAXagliptin: (Moderate) Monitor blood glucose during concomitant dapagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant saxagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.

Daratumumab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Deferasirox: (Moderate) Because gastric ulceration and GI bleeding have been reported in patients taking deferasirox, use caution when coadministering with other drugs known to increase the risk of peptic ulcers or gastric hemorrhage including salicylates.
Defibrotide: (Contraindicated) Coadministration of defibrotide with antithrombotic agents like aspirin is contraindicated. The pharmacodynamic activity and risk of hemorrhage with antithrombotic agents are increased if coadministered with defibrotide. If therapy with defibrotide is necessary, discontinue antithrombotic agents prior to initiation of defibrotide therapy. Consider delaying the onset of defibrotide treatment until the effects of the antithrombotic agent have abated.
Deflazacort: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Desvenlafaxine: (Moderate) Platelet aggregation may be impaired by serotonin norepinephrine reuptake inhibitors (SNRIs) due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin, ASA or other salicylates which affect hemostasis. Patients should be instructed to monitor for signs and symptoms of bleeding while taking an SNRI with medications which impair platelet function and to promptly report any bleeding events to the practitioner.
dexAMETHasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Dichlorphenamide: (Major) Dichlorphenamide is contraindicated with the concomitant use of high dose aspirin, ASA and should be used cautiously in patients receiving low dose aspirin. Dichlorphenamide may cause an elevation in salicylate concentrations in patients receiving aspirin. Adverse reactions including anorexia, tachypnea, lethargy, and coma have been reported with the concomitant use of dichlorphenamide and high dose aspirin.
Diclofenac: (Major) Concomitant use of analgesic doses of aspirin and diclofenac is

generally not recommended due to the increased risk of bleeding and renal impairment. Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Diclofenac is not a substitute for low dose aspirin for cardiovascular protection.
Diclofenac; miSOPROStol: (Major) Concomitant use of analgesic doses of aspirin and diclofenac is generally not recommended due to the increased risk of bleeding and renal impairment. Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Diclofenac is not a substitute for low dose aspirin for cardiovascular protection.
Dicloxacillin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Diflunisal: (Major) The concurrent use of diflunisal and salicylates is not recommended due to the increased risk of gastrointestinal toxicity with little or no increase in anti-inflammatory efficacy.
diphenhydrAMINE; Ibuprofen: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6

days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen 400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.
diphenhydrAMINE; Naproxen: (Major) Concomitant use of analgesic doses of aspirin and naproxen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events after discontinuation of naproxen due to the interference with the antiplatelet effect of aspirin during the washout period, for patients taking low-dose aspirin for cardioprotection who require intermittent analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics as appropriate. A pharmacodynamic study demonstrated that lower dose naproxen (220mg/day or 220mg twice daily) interfered with the antiplatelet effect of low-dose immediate-release aspirin, with the interaction most marked during the washout period of naproxen. There is reason to expect that the interaction would be present with prescription doses of naproxen or with enteric-coated low-dose aspirin; however, the peak interference with aspirin function may be later than observed in the study due to the longer washout period. A decrease in antiplatelet activity was observed at 24 hours after 10 days of naproxen 220 mg/day with low-dose immediate-release aspirin 81 mg/day (93.1%) vs. aspirin alone (98.7%). The interaction was observed even after discontinuation of naproxen on day 11 while aspirin therapy continued but normalized by day 13. The interaction was greater when naproxen was given 30 minutes before aspirin (87.7% vs. 98.7%) and minimal when aspirin was administered 30 minutes before naproxen (95.4% vs. 98.7%). The interaction was minimal at 24 hours after day 10 when naproxen 220 mg twice daily was given 30 minutes before low-dose immediate-release aspirin (95.7% vs. 98.7%); however, the interaction was greater on day 11 after naproxen discontinuation (84.3% vs. 98.7%) and did not normalize by day 13 (90.7% vs. 98.5%).

Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Naproxen is not a substitute for low dose aspirin for cardiovascular protection.
Dipyridamole: (Moderate) Monitor for bleeding during concomitant use of aspirin and dipyridamole; both medications interfere with platelet aggregation and concomitant use increases the risk for bleeding. For patients undergoing a dipyridamole nuclear stress test and receiving combination aspirin; dipyridamole, hold aspirin; dipyridamole for 48 hours prior to the stress test; concomitant use may increase the risk for cardiovascular side effects and impair the sensitivity of the test.
Doravirine; lamiVUDine; Tenofovir disoproxil fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Dorzolamide; Timolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Dulaglutide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
DULoxetine: (Moderate) Platelet aggregation may be impaired by serotonin norepinephrine reuptake inhibitors (SNRIs) due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin, ASA or other salicylates which affect hemostasis. Patients should be instructed to monitor for signs and symptoms of bleeding while taking an SNRI with medications which impair platelet function and to promptly report any bleeding events to the practitioner.
Edoxaban: (Major) Monitor for bleeding in patients who require chronic treatment with aspirin. Concomitant use of edoxaban with drugs that affect hemostasis, such as aspirin, may increase the risk of bleeding. The coadministration of aspirin (100 mg or 325 mg) and edoxaban increased bleeding time relative to that seen with either drug alone.
Efavirenz; Emtricitabine; Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have

underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Efavirenz; lamiVUDine; Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Efgartigimod Alfa; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Elvitegravir; Cobicistat; Emtricitabine; Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Empagliflozin: (Moderate) Monitor blood glucose during concomitant empagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Empagliflozin; Linagliptin: (Moderate) Monitor blood glucose during concomitant empagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant linagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Empagliflozin; Linagliptin; metFORMIN: (Moderate) Monitor blood glucose during concomitant empagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant linagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Empagliflozin; metFORMIN: (Moderate) Monitor blood glucose during concomitant empagliflozin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose

during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Emtricitabine; Rilpivirine; Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Emtricitabine; Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Enalapril, Enalaprilat: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Enalapril; hydroCHLOROthiazide, HCTZ: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Enoxaparin: (Moderate) An additive risk of bleeding may be seen in patients receiving a low molecular weight heparin in combination with other agents known to increase the

risk of bleeding such as salicylates. Monitor clinical and laboratory response closely during concurrent use.
Epoprostenol: (Moderate) When used concurrently with platelet inhibitors, epoprostenol may increase the risk of bleeding.
Eptifibatide: (Moderate) Unless contraindicated, aspirin is used in combination with eptifibatide. However, both drugs are associated with bleeding. Monitor for bleeding during concomitant therapy.
Ergotamine; Caffeine: (Minor) Caffeine has been reported to increase the metabolism of aspirin.
Ertugliflozin; metFORMIN: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Ertugliflozin; SITagliptin: (Moderate) Monitor blood glucose during concomitant sitagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Escitalopram: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Esmolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Ethacrynic Acid: (Moderate) Salicylates may decrease the diuretic, natriuretic, and antihypertensive actions of diuretics, possibly through inhibition of renal prostaglandin synthesis. Patients receiving loop diuretics and salicylates should be monitored for changes in the effectiveness of their diuretic therapy.
Ethanol: (Major) Concomitant ingestion of alcohol with salicylates, especially aspirin, ASA, increases the risk of developing gastric irritation and GI mucosal bleeding. Alcohol and salicylates are mucosal irritants and aspirin decreases platelet aggregation. Routine ingestion of alcohol and aspirin can cause significant GI bleeding, which may or may not be overt. Even occasional concomitant use of salicylates and alcohol should be avoided. Chronic ingestion of alcohol is often associated with hypoprothrombinemia and this condition increases the risk of salicylate-induced bleeding. Patients should be warned regarding the potential for increased risk of GI bleeding if alcohol-containing beverages are taken concurrently with salicylates.
Etodolac: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Exenatide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect

with risk of hypoglycemia.
Fenoprofen: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Fish Oil, Omega-3 Fatty Acids (Dietary Supplements): (Moderate) Because fish oil, omega-3 fatty acids inhibit platelet aggregation, caution is advised when fish oils are used concurrently with aspirin. Theoretically, the risk of bleeding may be increased.
Fludrocortisone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Flunisolide: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
FLUoxetine: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Flurbiprofen: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Fluticasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Fluticasone; Salmeterol: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Fluticasone; Umeclidinium; Vilanterol: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk

of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Fluticasone; Vilanterol: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
fluvoxaMINE: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Fondaparinux: (Moderate) An additive risk of bleeding may be seen in patients receiving platelet inhibitors (e.g., aspirin, ASA) in combination with fondaparinux. Data on the concomitant use of fondaparinux with aspirin are lacking; however, large doses of salicylates (>= 3-4 g/day) can cause hypoprothrombinemia, an additional risk factor for bleeding.
Formoterol; Mometasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Foscarnet: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents, such as foscarnet, may lead to additive nephrotoxicity.
Fosinopril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Fosinopril; hydroCHLOROthiazide, HCTZ: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor

in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Fosphenytoin: (Minor) Large doses of salicylates can displace phenytoin from plasma protein-binding sites. Although increased serum concentrations of unbound phenytoin may lead to phenytoin toxicity, the liver may also more rapidly clear unbound drug. Fosphenytoin is converted to phenytoin in vivo, so this interaction may also occur with fosphenytoin.
Furosemide: (Moderate) Salicylates may decrease the diuretic, natriuretic, and antihypertensive actions of diuretics, possibly through inhibition of renal prostaglandin synthesis. Patients receiving loop diuretics and salicylates should be monitored for changes in the effectiveness of their diuretic therapy.
Garlic, Allium sativum: (Moderate) Garlic, Allium sativum may produce clinically-significant antiplatelet effects; until more data are available, garlic should be used cautiously in patients receiving drugs with a potential risk for bleeding such as aspirin, ASA.
Gentamicin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
Ginger, Zingiber officinale: (Moderate) There may be an increased risk of bleeding in patients on aspirin therapy who take ginger as a supplement (i.e., usual dietary intake is not expected to pose a risk). Several pungent constituents of ginger, Zingiber officinale are reported to inhibit arachidonic acid induced platelet activation in human whole blood. Ginger-associated platelet inhibition may be related to a decrease in COX-1/Thromboxane synthase enzymatic activity. The increased risk of bleeding is theoretical; clinical data of an interaction are not available.
Ginkgo, Ginkgo biloba: (Moderate) Monitor for signs or symptoms of bleeding with coadministration of ginkgo biloba and aspirin as an increased bleeding risk may occur. Although data are mixed, ginkgo biloba is reported to inhibit platelet aggregation and several case reports describe bleeding complications with ginkgo biloba, with or without concomitant drug therapy.
Glimepiride: (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with

risk of hypoglycemia.
glipiZIDE: (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
glipiZIDE; metFORMIN: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
glyBURIDE: (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
glyBURIDE; metFORMIN: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Green Tea: (Moderate) Green tea should be used cautiously in patients taking aspirin; there may be an increased risk of bleeding. Monitoring clinical and/or laboratory parameters is warranted. Green tea has demonstrated antiplatelet and fibrinolytic actions in animals.
Griseofulvin: (Moderate) Concurrent administration of griseofulvin with salicylates may result in decreased salicylate serum concentrations. Caution and close monitoring for changes in the effectiveness of the salicylate are recommended.
Heparin: (Moderate) An additive risk of bleeding may be seen in patients receiving platelet inhibitors (e.g., aspirin). Despite the potential drug-drug interaction between aspirin and heparin, heparin is frequently administered in combination with low-dose aspirin therapy to patients who have had an acute myocardial infarction and in other disease states. In addition, large doses of salicylates (at least 3 to 4 g/day) can cause hypoprothrombinemia, an additional risk factor for bleeding.
Hyaluronidase, Recombinant; Immune Globulin: (Moderate) Immune Globulin (IG) products have been reported to be associated with renal dysfunction, acute renal failure, osmotic nephrosis, and death. Patients predisposed to acute renal failure include patients receiving known nephrotoxic drugs like nonsteroidal anti-inflammatory drugs (NSAIDs) and salicylates. Coadminister IG products at the minimum concentration available and the minimum rate of infusion practicable. Also, closely monitor renal function. (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render

tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
hydroCHLOROthiazide, HCTZ; Moexipril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
HYDROcodone; Ibuprofen: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily

was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6 days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen 400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.
Hydrocortisone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Hyoscyamine; Methenamine; Methylene Blue; Phenyl Salicylate; Sodium Biphosphate: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Ibandronate: (Moderate) Monitor renal function and for gastrointestinal adverse events during concurrent use of intravenous or oral ibandronate use, respectively, and aspirin. Acute renal failure has been observed with intravenous ibandronate and concomitant use of other nephrotoxic agents may increase this risk. Additionally, the oral formulations of both medications have been associated with gastrointestinal irritation although data suggest concomitant use introduces little additional risk for adverse effects for most patients.
Ibritumomab Tiuxetan: (Major) During and after therapy, avoid the concomitant use of Yttrium (Y)-90 ibrutumomab tiuxetan with drugs that interfere with platelet function such as aspirin; the risk of bleeding may be increased. If coadministration with asprin is necessary, monitor platelet counts more frequently for evidence of thrombocytopenia.
(Moderate) Acidification of the urine may increase serum concentrations of salicylates by

increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic. (Moderate) Agents that acidify the urine, like phosphate salts, should be avoided in patients receiving high-dose salicylates. Urine acidifying agents may increase renal tubular reabsorption of salicylic acid and possibly increase salicylic acid levels. (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Ibrutinib: (Moderate) The concomitant use of ibrutinib and antiplatelet agents such as aspirin may increase the risk of bleeding; monitor patients for signs of bleeding. Severe bleeding events have occurred with ibrutinib therapy including intracranial hemorrhage, GI bleeding, hematuria, and post procedural hemorrhage; some events were fatal. The mechanism for bleeding with ibrutinib therapy is not well understood. Also, aspirin may mask signs of infection such as fever and in patients following treatment with antineoplastic agents or immunosuppressives.
Ibuprofen: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6 days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen

400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.
Ibuprofen; Famotidine: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6 days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen 400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as

compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.
Ibuprofen; Pseudoephedrine: (Major) Concomitant use of analgesic doses of aspirin and ibuprofen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events due to the interference of ibuprofen with the antiplatelet effect of aspirin, for patients taking low-dose aspirin for cardioprotection who require analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics, as appropriate. Administer single doses of ibuprofen at least 2 to 4 hours or more after aspirin and wait 8 hours after ibuprofen administration before administering aspirin to avoid significant interference. Monitor for signs and symptoms of renal impairment. Pharmacodynamic studies have demonstrated interference with the antiplatelet activity of aspirin when ibuprofen 400 mg 3 times daily is administered with enteric-coated low-dose aspirin. The interaction exists even after ibuprofen 400 mg once daily, particularly when ibuprofen is dosed prior to aspirin. The interaction is alleviated if immediate-release low-dose aspirin is dosed at least 2 hours prior to a once daily regimen of ibuprofen; however, this finding cannot be extended to enteric-coated low-dose aspirin. A decrease in antiplatelet activity (53%) was observed when ibuprofen 400 mg once daily was administered 2 hours before low-dose immediate-release aspirin 81 mg/day for 6 days. An interaction was still observed, but minimized, when ibuprofen 400 mg once daily was administered as early as 8 hours before immediate-release aspirin (90.7%). There was no interaction with the antiplatelet activity of aspirin when ibuprofen 400 mg once daily was administered 2 hours after immediate-release aspirin (99.2%). In another study of low-dose immediate-release aspirin 81 mg/day and ibuprofen 400 mg 3 times daily (1, 7, and 13 hours post-aspirin dose) for 10 consecutive days, there was no interaction with the antiplatelet activity of aspirin (98.3%); however, there were individuals with aspirin antiplatelet activity below 95%, with the lowest being 90.2%. When a similarly designed study was conducted with enteric-coated aspirin 81 mg/day for 6 days and ibuprofen 400 mg 3 times daily (2, 7 and 12 h post-aspirin dose) for 6 days, there was an interaction with the antiplatelet activity at 24 hours after the day 6 aspirin dose (67%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Ibuprofen is not a substitute for low dose aspirin for cardiovascular protection.
Iloprost: (Moderate) When used concurrently with platelet inhibitors, inhaled iloprost may increase the risk of bleeding.
Immune Globulin (IVIG, IMIG, SCIG): (Moderate) Immune Globulin (IG) products have been reported to be associated with renal dysfunction, acute renal failure, osmotic

nephrosis, and death. Patients predisposed to acute renal failure include patients receiving known nephrotoxic drugs like nonsteroidal anti-inflammatory drugs (NSAIDs) and salicylates. Coadminister IG products at the minimum concentration available and the minimum rate of infusion practicable. Also, closely monitor renal function.
Incretin Mimetics: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Indapamide: (Moderate) Salicylates can increase the risk of renal toxicity in patients receiving diuretics because salicylates inhibit renal prostaglandin synthesis, which can lead to fluid retention and increased peripheral vascular resistance.
Indomethacin: (Major) The concurrent use of salicylates and indomethacin is not recommended. Combined use does not produce any greater therapeutic effect than indomethacin monotherapy. Also, a significantly greater incidence of gastrointestinal adverse effects with concurrent use has been observed. Because NSAIDs can cause GI bleeding, inhibit platelet aggregation, and prolong bleeding time, additive effects may be seen in patients receiving platelet inhibitors (e.g., aspirin), anticoagulants, or thrombolytic agents.
Inotersen: (Moderate) Use caution with concomitant use of inotersen and salicylates due to the risk of glomerulonephritis and nephrotoxicity as well as the potential risk of bleeding from thrombocytopenia. Consider discontinuation of salicylates in a patient taking inotersen with a platelet count of less than 50,000 per microliter.
Insulin Aspart: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Aspart; Insulin Aspart Protamine: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Degludec: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Degludec; Liraglutide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Detemir: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Glargine: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect

with risk of hypoglycemia.
Insulin Glargine; Lixisenatide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Glulisine: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Lispro: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin Lispro; Insulin Lispro Protamine: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulin, Inhaled: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Insulins: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Intravenous Lipid Emulsions: (Moderate) Because fish oil, omega-3 fatty acids inhibit platelet aggregation, caution is advised when fish oils are used concurrently with aspirin. Theoretically, the risk of bleeding may be increased.
Irbesartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Iron Sucrose, Sucroferric Oxyhydroxide: (Moderate) Administer aspirin at least 1 hour before oral iron sucrose, sucroferric oxyhydroxide. Oral iron salts may reduce the bioavailability of aspirin, leading to decreased absorption.
Isophane Insulin (NPH): (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Ketoprofen: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.

Ketorolac: (Contraindicated) Ketorolac is contraindicated in patients currently receiving salicylates due to increased risk of serious NSAID-related adverse events, including gastrointestinal bleeding, ulceration, and perforation.
Labetalol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
lamiVUDine; Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Landiolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Lansoprazole; Amoxicillin; Clarithromycin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Levobunolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Levomilnacipran: (Moderate) Platelet aggregation may be impaired by serotonin norepinephrine reuptake inhibitors (SNRIs) due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin, ASA or other salicylates which affect hemostasis. Patients should be instructed to monitor for signs and symptoms of bleeding while taking an SNRI with medications which impair platelet function and to promptly report any bleeding events to the practitioner.
Linagliptin: (Moderate) Monitor blood glucose during concomitant linagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Linagliptin; metFORMIN: (Moderate) Monitor blood glucose during concomitant linagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.

Liraglutide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Lisinopril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Lisinopril; hydroCHLOROthiazide, HCTZ: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Lixisenatide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Loop diuretics: (Moderate) Salicylates may decrease the diuretic, natriuretic, and antihypertensive actions of diuretics, possibly through inhibition of renal prostaglandin synthesis. Patients receiving loop diuretics and salicylates should be monitored for changes in the effectiveness of their diuretic therapy.
Losartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Low Molecular Weight Heparins: (Moderate) An additive risk of bleeding may be seen in

patients receiving a low molecular weight heparin in combination with other agents known to increase the risk of bleeding such as salicylates. Monitor clinical and laboratory response closely during concurrent use.
Macimorelin: (Major) Avoid use of macimorelin with drugs that directly affect pituitary growth hormone secretion, such as salicylates. Healthcare providers are advised to discontinue salicylate therapy and observe a sufficient washout period before administering macimorelin. Use of these medications together may impact the accuracy of the macimorelin growth hormone test.
Mafenide: (Minor) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as sulfonamides. An enhanced effect of the displaced drug may occur.
Mannitol: (Major) In general, avoid use of mannitol and salicylates. Concomitant administration of nephrotoxic drugs, such as the salicylates, increases the risk of renal failure after administration of mannitol. However, mannitol promotes the urinary excretion of salicylates, and may be used as an adjunct in salicylate intoxication.
Measles Virus; Mumps Virus; Rubella Virus; Varicella Virus Vaccine, Live: (Major) No adverse events associated with the use of salicylates after varicella vaccination have been reported. However, the manufacturer of varicella virus vaccine live recommends the avoidance of salicylates or aspirin, ASA use for 6 weeks after vaccination. Reye's syndrome, which exclusively affects children under 15 years old, has been associated with aspirin use following active varicella infection. Vaccination with close clinical monitoring is recommended for children who require therapeutic aspirin, ASA therapy; according to the CDC the use of attenuated, live varicella virus vaccine is thought to present less risk than natural varicella disease to such children.
Meclofenamate Sodium: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Mefenamic Acid: (Major) Increased adverse gastrointestinal (GI) effects are possible if mefenamic acid is used with salicylates. In addition, concomitant administration of salicylates and mefenamic acid may result in an increase in unbound plasma concentrations of either drug, which could result in greater adverse effects. In general, concomitant use of aspirin and mefenamic acid is not recommended.
Meglitinides: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood sugar and may potentiate the effects of antidiabetic agents. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose or use of greater than maximum recommended daily dosages, salicylates can cause either hypoglycemia or hyperglycemia. Large doses of aspirin should be used cautiously in patients who receive antidiabetic agents.

Meloxicam: (Major) Concomitant use of low dose aspirin or analgesic doses of aspirin and meloxicam is generally not recommended due to the increased risk of bleeding and renal impairment. Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Meloxicam is not a substitute for low dose aspirin for cardiovascular protection.
Meloxicam; Rizatriptan: (Major) Concomitant use of low dose aspirin or analgesic doses of aspirin and meloxicam is generally not recommended due to the increased risk of bleeding and renal impairment. Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Meloxicam is not a substitute for low dose aspirin for cardiovascular protection.
metFORMIN: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
metFORMIN; sAXagliptin: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant saxagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
metFORMIN; SITagliptin: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant sitagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
methazolAMIDE: (Major) Avoid the coadministration of high-dose salicylates and carbonic anhydrase inhibitors, like methazolamide, whenever possible. The combination yielded reports of anorexia, tachypnea, lethargy, metabolic acidosis, coma, and death. The mechanism appears to be accumulation of the carbonic anhydrase inhibitor, resulting in increased CNS depression and metabolic acidosis. The acidosis may allow greater CNS penetration of the salicylate.
Methenamine; Sodium Acid Phosphate; Methylene Blue; Hyoscyamine: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Methohexital: (Moderate) Due to high protein binding, salicylates could be displaced

from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Methotrexate: (Major) Do not administer salicylates before or concomitantly with high doses of methotrexate, such as used in the treatment of osteosarcoma. Concomitant administration of some NSAIDs with high dose methotrexate therapy has been reported to elevate and prolong serum methotrexate concentrations, resulting in deaths from severe hematologic and gastrointestinal toxicity. Use caution when salicylates are administered concomitantly with lower doses of methotrexate. Salicylates have been reported to reduce the tubular secretion of methotrexate in an animal model and may enhance its toxicity. Methotrexate is partially bound to serum albumin, and toxicity may be increased because of displacement by salicylates.
methylPREDNISolone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Metoclopramide: (Minor) Metoclopramide can increase the rate or extent of absorption of aspirin because of accelerated gastric emptying, which increases the contact time with the small bowel where this drug is absorbed.
metOLazone: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Metoprolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Metoprolol; hydroCHLOROthiazide, HCTZ: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.

Miglitol: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood sugar. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Milnacipran: (Moderate) Platelet aggregation may be impaired by serotonin norepinephrine reuptake inhibitors (SNRIs) due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin, ASA or other salicylates which affect hemostasis. Patients should be instructed to monitor for signs and symptoms of bleeding while taking an SNRI with medications which impair platelet function and to promptly report any bleeding events to the practitioner.
Moexipril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Mometasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Mycophenolate: (Moderate) Mycophenolic acid is more than 98% bound to albumin. Concurrent use of mycophenolate with salicylates can decrease the protein binding of mycophenolic acid resulting in an increase in the free fraction of MPA. Patients should be observed for increased clinical effects from mycophenolate as well as additive adverse effects.
Nabumetone: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Nadolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Nafcillin: (Minor) Due to the high protein binding of aspirin, it could displace or be

displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Naproxen: (Major) Concomitant use of analgesic doses of aspirin and naproxen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events after discontinuation of naproxen due to the interference with the antiplatelet effect of aspirin during the washout period, for patients taking low-dose aspirin for cardioprotection who require intermittent analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics as appropriate. A pharmacodynamic study demonstrated that lower dose naproxen (220mg/day or 220mg twice daily) interfered with the antiplatelet effect of low-dose immediate-release aspirin, with the interaction most marked during the washout period of naproxen. There is reason to expect that the interaction would be present with prescription doses of naproxen or with enteric-coated low-dose aspirin; however, the peak interference with aspirin function may be later than observed in the study due to the longer washout period. A decrease in antiplatelet activity was observed at 24 hours after 10 days of naproxen 220 mg/day with low-dose immediate-release aspirin 81 mg/day (93.1%) vs. aspirin alone (98.7%). The interaction was observed even after discontinuation of naproxen on day 11 while aspirin therapy continued but normalized by day 13. The interaction was greater when naproxen was given 30 minutes before aspirin (87.7% vs. 98.7%) and minimal when aspirin was administered 30 minutes before naproxen (95.4% vs. 98.7%). The interaction was minimal at 24 hours after day 10 when naproxen 220 mg twice daily was given 30 minutes before low-dose immediate-release aspirin (95.7% vs. 98.7%); however, the interaction was greater on day 11 after naproxen discontinuation (84.3% vs. 98.7%) and did not normalize by day 13 (90.7% vs. 98.5%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Naproxen is not a substitute for low dose aspirin for cardiovascular protection.
Naproxen; Esomeprazole: (Major) Concomitant use of analgesic doses of aspirin and naproxen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events after discontinuation of naproxen due to the interference with the antiplatelet effect of aspirin during the washout period, for patients taking low-dose aspirin for cardioprotection who require intermittent analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics as appropriate. A pharmacodynamic study demonstrated that lower dose naproxen

(220mg/day or 220mg twice daily) interfered with the antiplatelet effect of low-dose immediate-release aspirin, with the interaction most marked during the washout period of naproxen. There is reason to expect that the interaction would be present with prescription doses of naproxen or with enteric-coated low-dose aspirin; however, the peak interference with aspirin function may be later than observed in the study due to the longer washout period. A decrease in antiplatelet activity was observed at 24 hours after 10 days of naproxen 220 mg/day with low-dose immediate-release aspirin 81 mg/day (93.1%) vs. aspirin alone (98.7%). The interaction was observed even after discontinuation of naproxen on day 11 while aspirin therapy continued but normalized by day 13. The interaction was greater when naproxen was given 30 minutes before aspirin (87.7% vs. 98.7%) and minimal when aspirin was administered 30 minutes before naproxen (95.4% vs. 98.7%). The interaction was minimal at 24 hours after day 10 when naproxen 220 mg twice daily was given 30 minutes before low-dose immediate-release aspirin (95.7% vs. 98.7%); however, the interaction was greater on day 11 after naproxen discontinuation (84.3% vs. 98.7%) and did not normalize by day 13 (90.7% vs. 98.5%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Naproxen is not a substitute for low dose aspirin for cardiovascular protection.
Naproxen; Pseudoephedrine: (Major) Concomitant use of analgesic doses of aspirin and naproxen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events after discontinuation of naproxen due to the interference with the antiplatelet effect of aspirin during the washout period, for patients taking low-dose aspirin for cardioprotection who require intermittent analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics as appropriate. A pharmacodynamic study demonstrated that lower dose naproxen (220mg/day or 220mg twice daily) interfered with the antiplatelet effect of low-dose immediate-release aspirin, with the interaction most marked during the washout period of naproxen. There is reason to expect that the interaction would be present with prescription doses of naproxen or with enteric-coated low-dose aspirin; however, the peak interference with aspirin function may be later than observed in the study due to the longer washout period. A decrease in antiplatelet activity was observed at 24 hours after 10 days of naproxen 220 mg/day with low-dose immediate-release aspirin 81 mg/day (93.1%) vs. aspirin alone (98.7%). The interaction was observed even after discontinuation of naproxen on day 11 while aspirin therapy continued but normalized by day 13. The interaction was greater when naproxen was given 30 minutes before aspirin (87.7% vs. 98.7%) and minimal when aspirin was administered 30 minutes before

naproxen (95.4% vs. 98.7%). The interaction was minimal at 24 hours after day 10 when naproxen 220 mg twice daily was given 30 minutes before low-dose immediate-release aspirin (95.7% vs. 98.7%); however, the interaction was greater on day 11 after naproxen discontinuation (84.3% vs. 98.7%) and did not normalize by day 13 (90.7% vs. 98.5%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Naproxen is not a substitute for low dose aspirin for cardiovascular protection.
Nateglinide: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood sugar and may potentiate the effects of antidiabetic agents. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose or use of greater than maximum recommended daily dosages, salicylates can cause either hypoglycemia or hyperglycemia. Large doses of aspirin should be used cautiously in patients who receive antidiabetic agents.
Nebivolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Nitazoxanide: (Moderate) The active metabolite of nitazoxanide, tizoxanide, is highly bound to plasma proteins. Caution should be exercised when administering nitazoxanide concurrently with other highly plasma protein-bound drugs with narrow therapeutic indices because competition for binding sites may occur.
Nitroglycerin: (Moderate) When coadministered with aspirin, ASA (doses between 500 mg and 1000 mg), the maximum plasma concentration (Cmax) and exposure (AUC) of a single nitroglycerin dose is increased by 67% and 73%, respectively. Additionally, limited data suggest that patients receiving aspirin, ASA in high doses can exhibit an exaggerated response to sublingual nitroglycerin. Although hypotension and tachycardia were more significant during concomitant therapy, no special precautions appear necessary. The pharmacologic effects of 0.4% nitroglycerin rectal ointment may also be enhanced when administered concomitantly with aspirin, ASA; therefore, close clinical monitoring is advised.
Nivolumab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Ocrelizumab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving

these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
OLANZapine; FLUoxetine: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Olmesartan; amLODIPine; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Olmesartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Olopatadine; Mometasone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Omacetaxine: (Major) Avoid the concomitant use of omacetaxine and aspirin, ASA when the platelet count is less than 50,000 cells/microliter due to an increased risk of bleeding. Also, aspirin may mask signs of infection such as fever and pain in patients following treatment with antineoplastic agents or immunosuppressives. Aspirin, ASA should be used with caution in patients receiving immunosuppressive therapy. Special consideration should be given to myelosuppressed patients prior to receiving aspirin.
Omeprazole; Amoxicillin; Rifabutin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Omeprazole; Sodium Bicarbonate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by

decreasing renal tubular reabsorption of salicylic acid.
Oxacillin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Oxaprozin: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Pamidronate: (Moderate) Monitor renal function during concomitant pamidronate and aspirin use due to risk for additive nephrotoxicity.
Paromomycin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
PARoxetine: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Pembrolizumab; berahyaluronidase alfa: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Penicillin G Benzathine: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Penicillin G Benzathine; Penicillin G Procaine: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Penicillin G Procaine: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Penicillin G: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin

serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Penicillin V: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Penicillins: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
PENTobarbital: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Pentosan: (Moderate) Pentosan is a weak anticoagulant. Pentosan has 1/15 the anticoagulant activity of heparin. An additive risk of bleeding may be seen in patients receiving other platelet inhibitors (e.g., aspirin, ASA) in combination with pentosan. Also, large doses of salicylates (>= 3-4 g/day) can cause hypoprothrombinemia, an additional risk factor for bleeding.
Pentoxifylline: (Moderate) The concomitant administration of platelet inhibitor with pentoxifylline in the treatment of intermittent claudication has not been evaluated and should be approached with caution, due to the potential for synergistic effects.
Perindopril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Perindopril; amLODIPine: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Pertuzumab; Trastuzumab; Hyaluronidase: (Minor) Salicylates, when given in large

systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
PHENobarbital: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
PHENobarbital; Hyoscyamine; Atropine; Scopolamine: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Phenytoin: (Minor) Large doses of salicylates can displace phenytoin from plasma protein-binding sites. Although increased serum concentrations of unbound phenytoin may lead to phenytoin toxicity, the liver may also more rapidly clear unbound drug. Displacement of phenytoin from binding sites can lead to a decrease in the total phenytoin serum concentration. Close monitoring for excessive phenytoin toxicity or decreased phenytoin efficacy is recommended.
Phosphorated Carbohydrate Solution: (Moderate) Agents that acidify the urine, like phosphate salts, should be avoided in patients receiving high-dose salicylates. Urine acidifying agents may increase renal tubular reabsorption of salicylic acid and possibly increase salicylic acid levels.
Phosphorus: (Moderate) Agents that acidify the urine, like phosphate salts, should be avoided in patients receiving high-dose salicylates. Urine acidifying agents may increase renal tubular reabsorption of salicylic acid and possibly increase salicylic acid levels.
Photosensitizing agents (topical): (Minor) Preclinical data suggest that agents that affect platelet function and inhibit prostaglandin synthesis could decrease the efficacy of photosensitizing agents used during photodynamic therapy.
Pindolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Pioglitazone: (Moderate) Monitor blood glucose during concomitant thiazolidinedione and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Pioglitazone; Glimepiride: (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during concomitant thiazolidinedione and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Pioglitazone; metFORMIN: (Moderate) Monitor blood glucose during concomitant metformin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia. (Moderate) Monitor blood glucose during

concomitant thiazolidinedione and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Piperacillin; Tazobactam: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Piroxicam: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Pivmecillinam: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Plazomicin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
Pneumococcal Vaccine, Polyvalent: (Moderate) Concomitant administration of antipyretics, such as aspirin, ASA, may decrease an individual's immunological response to the pneumococcal vaccine. A post-marketing study conducted in Poland using a non-US vaccination schedule (2, 3, 4, and 12 months of age) evaluated the impact of prophylactic oral acetaminophen on antibody responses to Prevnar 13. Data show that acetaminophen, given at the time of vaccination and then dosed at 6 to 8 hour intervals for 3 doses on a scheduled basis, reduced the antibody response to some serotypes after the third dose of Prevnar 13 when compared to the antibody responses of infants who only received antipyretics 'as needed' for treatment. However, reduced antibody responses were not observed after the fourth dose of Prevnar 13 with prophylactic acetaminophen.
Potassium Bicarbonate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid. (Moderate) Urinary alkalinizing agents, like potassium citrate, increase the excretion of salicylates by increasing renal clearance.
Potassium Chloride: (Moderate) Urinary alkalinizing agents, like potassium citrate, increase the excretion of salicylates by increasing renal clearance.
Potassium Citrate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid. (Moderate) Urinary alkalinizing agents, like potassium citrate, increase the excretion of salicylates by increasing renal clearance.

Potassium Citrate; Citric Acid: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid. (Moderate) Urinary alkalinizing agents, like potassium citrate, increase the excretion of salicylates by increasing renal clearance.
Potassium Phosphate: (Moderate) Agents that acidify the urine, like phosphate salts, should be avoided in patients receiving high-dose salicylates. Urine acidifying agents may increase renal tubular reabsorption of salicylic acid and possibly increase salicylic acid levels.
Potassium Phosphate; Sodium Phosphate: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic. (Moderate) Agents that acidify the urine, like phosphate salts, should be avoided in patients receiving high-dose salicylates. Urine acidifying agents may increase renal tubular reabsorption of salicylic acid and possibly increase salicylic acid levels.
Pramlintide: (Moderate) Salicylates can indirectly increase insulin secretion, and thus decrease blood glucose concentrations. In large doses, salicylates may cause hyperglycemia and glycosuria. After acute overdose, aspirin can cause either hypo- or hyperglycemia. Large doses of aspirin should be used cautiously in patients receiving antidiabetic agents.
Prasterone, Dehydroepiandrosterone, DHEA (Dietary Supplements): (Moderate) Prasterone, dehydroepiandrosterone, DHEA appears to have antiplatelet effects, which may prolong bleeding times. Because of these potential, varied effects on coagulation, patients receiving DHEA concurrently with aspirin, should be monitored for side effects or the need for dosage adjustments.
Prasterone, Dehydroepiandrosterone, DHEA (FDA-approved): (Moderate) Prasterone, dehydroepiandrosterone, DHEA appears to have antiplatelet effects, which may prolong bleeding times. Because of these potential, varied effects on coagulation, patients receiving DHEA concurrently with aspirin, should be monitored for side effects or the need for dosage adjustments.
Prasugrel: (Moderate) Although indicated for concomitant use, both prasugrel and aspirin are associated with bleeding. Aspirin 150 mg did not alter prasugrel-mediated inhibition of platelet aggregation; however, bleeding time was increased compared to either drug alone. Monitor for bleeding during concomitant therapy.
prednisoLONE: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of

renal clearance.
predniSONE: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Primidone: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Probenecid: (Contraindicated) Concurrent use of probenecid and salicylates is contraindicated. The uricosuric actions of probenecid are inhibited by salicylates. When probenecid is used to treat hyperuricemia or gout, do not administer with salicylates.
Probenecid; Colchicine: (Contraindicated) Concurrent use of probenecid and salicylates is contraindicated. The uricosuric actions of probenecid are inhibited by salicylates. When probenecid is used to treat hyperuricemia or gout, do not administer with salicylates.
Propranolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Psyllium: (Moderate) Psyllium can interfere with the absorption of certain oral drugs if administered concomitantly. For example, psyllium fiber can adsorb salicylates. Per the psyllium manufacturers, administration of other prescribed oral drugs should be separated from the administration of psyllium by at least 2 hours.
Quinapril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Quinapril; hydroCHLOROthiazide, HCTZ: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Moderate)

Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Ramipril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Regular Insulin: (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Regular Insulin; Isophane Insulin (NPH): (Moderate) Monitor blood glucose during concomitant insulin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Repaglinide: (Moderate) Salicylates, by inhibiting prostaglandin E2 synthesis, can indirectly increase insulin secretion. Thus, salicylates can decrease blood sugar and may potentiate the effects of antidiabetic agents. In large doses, salicylates uncouple oxidative phosphorylation, deplete hepatic and muscle glycogen, and cause hyperglycemia and glycosuria. After acute overdose or use of greater than maximum recommended daily dosages, salicylates can cause either hypoglycemia or hyperglycemia. Large doses of aspirin should be used cautiously in patients who receive antidiabetic agents.
Reteplase, r-PA: (Moderate) Monitor for bleeding if the concomitant use of thrombolytics and aspirin is necessary. For patients who received a thrombolytic for acute ischemic stroke, avoid IV aspirin in the first 90 minutes following thrombolytic administration and consider delaying the initiation of aspirin therapy for at least 24 hours following thrombolytic administration. Both medications interfere with hemostasis and concomitant use increases the risk for bleeding.
Risedronate: (Minor) Monitor for gastrointestinal adverse events during concurrent use of risedronate and aspirin. Both medications have been associated with gastrointestinal irritation although data suggest concomitant use introduces little additional risk for adverse effects for most patients.
riTUXimab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these

medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
Rivaroxaban: (Major) Salicylates such as aspirin are known to increase bleeding, and bleeding risk may be increased when these drugs are used concomitantly with rivaroxaban. The safety of long-term concomitant use of these drugs has not been studied. Promptly evaluate any signs or symptoms of bleeding or blood loss if patients are treated concomitantly with salicylates. In a single-dose drug interaction study, no pharmacokinetic interactions were observed after concomitant administration of acetylsalicylic acid (aspirin, ASA) with rivaroxaban.
Rosiglitazone: (Moderate) Monitor blood glucose during concomitant thiazolidinedione and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Salicylic Acid: (Moderate) Concomitant use of salicylic acid with other drugs which may contribute to elevated serum salicylate levels (e.g., oral aspirin or other oral salicylates and other salicylate containing medications, such as sports injury creams) should be avoided. Concurrent use may result in excessive exposure to salicylic acid. Consider replacing aspirin therapy with an alternative non-steroidal anti-inflammatory agent that is not salicylate based where appropriate.
sAXagliptin: (Moderate) Monitor blood glucose during concomitant saxagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Secobarbital: (Moderate) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as barbiturates. An enhanced effect of the displaced drug may occur.
Selective serotonin reuptake inhibitors: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
Semaglutide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Serotonin norepinephrine reuptake inhibitors: (Moderate) Platelet aggregation may be impaired by serotonin norepinephrine reuptake inhibitors (SNRIs) due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin, ASA or other salicylates which affect hemostasis. Patients should be instructed to monitor for signs and symptoms of bleeding while taking an SNRI with medications which impair platelet function and to promptly report any bleeding events to the practitioner.

Sertraline: (Moderate) The combined use of selective serotonin reuptake inhibitors and aspirin, ASA may elevate the risk for an upper GI bleed. SSRIs may inhibit serotonin uptake by platelets, augmenting the antiplatelet effects of aspirin. Additionally, aspirin impairs the gastric mucosa defenses by inhibiting prostaglandin formation.
SITagliptin: (Moderate) Monitor blood glucose during concomitant sitagliptin and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Sodium Acetate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Sodium Benzoate; Sodium Phenylacetate: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Sodium Bicarbonate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Sodium Citrate; Citric Acid: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Sodium Lactate: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Sodium Phosphate Monobasic Monohydrate; Sodium Phosphate Dibasic Anhydrous: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Sodium Thiosulfate; Salicylic Acid: (Moderate) Concomitant use of salicylic acid with other drugs which may contribute to elevated serum salicylate levels (e.g., oral aspirin or other oral salicylates and other salicylate containing medications, such as sports injury creams) should be avoided. Concurrent use may result in excessive exposure to salicylic acid. Consider replacing aspirin therapy with an alternative non-steroidal anti-inflammatory agent that is not salicylate based where appropriate.
Sotalol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Spironolactone: (Moderate) Monitor renal function and for decreased efficacy of spironolactone if coadministration with aspirin is necessary. The spironolactone dose may need to be titrated to higher maintenance dose. In persons who are elderly, volume-depleted (including those receiving diuretic therapy), or with compromised renal

function, coadministration of spironolactone and aspirin may result in deterioration of renal function, including possible acute renal failure; these effects are usually reversible. Aspirin may reduce the efficacy of spironolactone. A single aspirin 600 mg dose inhibited the natriuretic effect of spironolactone, which was hypothesized be due to inhibition of tubular secretion of canrenone, causing decreased effectiveness of spironolactone.
Spironolactone; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention. (Moderate) Monitor renal function and for decreased efficacy of spironolactone if coadministration with aspirin is necessary. The spironolactone dose may need to be titrated to higher maintenance dose. In persons who are elderly, volume-depleted (including those receiving diuretic therapy), or with compromised renal function, coadministration of spironolactone and aspirin may result in deterioration of renal function, including possible acute renal failure; these effects are usually reversible. Aspirin may reduce the efficacy of spironolactone. A single aspirin 600 mg dose inhibited the natriuretic effect of spironolactone, which was hypothesized be due to inhibition of tubular secretion of canrenone, causing decreased effectiveness of spironolactone.
Streptomycin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
Sulfacetamide; Sulfur: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
sulfADIAZINE: (Minor) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as sulfonamides. An enhanced effect of the displaced drug may occur.
Sulfamethoxazole; Trimethoprim, SMX-TMP, Cotrimoxazole: (Minor) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as sulfonamides. An enhanced effect of the displaced drug may occur.
sulfaSALAzine: (Minor) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as sulfonamides. An enhanced effect of the displaced drug may occur.
Sulfonamides: (Minor) Due to high protein binding, salicylates could be displaced from binding sites, or could displace other highly protein-bound drugs such as sulfonamides. An enhanced effect of the displaced drug may occur.

Sulfonylureas: (Moderate) Monitor blood glucose during concomitant sulfonylurea and aspirin use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Sulindac: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Sulopenem Etzadroxil; Probenecid: (Contraindicated) Concurrent use of probenecid and salicylates is contraindicated. The uricosuric actions of probenecid are inhibited by salicylates. When probenecid is used to treat hyperuricemia or gout, do not administer with salicylates.
SUMAtriptan; Naproxen: (Major) Concomitant use of analgesic doses of aspirin and naproxen is generally not recommended due to the increased risk of bleeding and renal impairment. Because there may be an increased risk of cardiovascular events after discontinuation of naproxen due to the interference with the antiplatelet effect of aspirin during the washout period, for patients taking low-dose aspirin for cardioprotection who require intermittent analgesics, consider use of an NSAID that does not interfere with the antiplatelet effect of aspirin, or non-NSAID analgesics as appropriate. A pharmacodynamic study demonstrated that lower dose naproxen (220mg/day or 220mg twice daily) interfered with the antiplatelet effect of low-dose immediate-release aspirin, with the interaction most marked during the washout period of naproxen. There is reason to expect that the interaction would be present with prescription doses of naproxen or with enteric-coated low-dose aspirin; however, the peak interference with aspirin function may be later than observed in the study due to the longer washout period. A decrease in antiplatelet activity was observed at 24 hours after 10 days of naproxen 220 mg/day with low-dose immediate-release aspirin 81 mg/day (93.1%) vs. aspirin alone (98.7%). The interaction was observed even after discontinuation of naproxen on day 11 while aspirin therapy continued but normalized by day 13. The interaction was greater when naproxen was given 30 minutes before aspirin (87.7% vs. 98.7%) and minimal when aspirin was administered 30 minutes before naproxen (95.4% vs. 98.7%). The interaction was minimal at 24 hours after day 10 when naproxen 220 mg twice daily was given 30 minutes before low-dose immediate-release aspirin (95.7% vs. 98.7%); however, the interaction was greater on day 11 after naproxen discontinuation (84.3% vs. 98.7%) and did not normalize by day 13 (90.7% vs. 98.5%). Controlled clinical studies showed that the concomitant use of NSAIDs and analgesic doses of aspirin does not produce any greater therapeutic effect than the use of NSAIDs alone. In a clinical study, the concomitant use of an NSAID and aspirin was associated with a significantly increased incidence of GI adverse reactions as compared to use of the NSAID alone. Naproxen is not a substitute for low dose aspirin for cardiovascular protection.
Tacrolimus: (Moderate) Tacrolimus, in the absence of overt renal impairment, may

adversely affect renal function. Care should be taken in using tacrolimus with other nephrotoxic drugs, such as salicylates.
Telavancin: (Minor) Concurrent or sequential use of telavancin with drugs that inhibit renal prostaglandins such as salicylates may lead to additive nephrotoxicity. Closely monitor renal function and adjust telavancin doses based on calculated creatinine clearance.
Telmisartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Tenecteplase: (Moderate) Monitor for bleeding if the concomitant use of thrombolytics and aspirin is necessary. For patients who received a thrombolytic for acute ischemic stroke, avoid IV aspirin in the first 90 minutes following thrombolytic administration and consider delaying the initiation of aspirin therapy for at least 24 hours following thrombolytic administration. Both medications interfere with hemostasis and concomitant use increases the risk for bleeding.
Tenofovir Disoproxil Fumarate: (Major) Renal impairment, which may include hypophosphatemia, has been reported with the use of tenofovir disoproxil fumarate with a majority of the cases occurring in patients who have underlying systemic or renal disease or who are concurrently taking nephrotoxic agents. Tenofovir should be avoided with concurrent or recent use of a nephrotoxic agent; patients receiving concomitant nephrotoxic agents, like salicylates should be carefully monitored for changes in serum creatinine and phosphorus.
Thiazide diuretics: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Thiazolidinediones: (Moderate) Monitor blood glucose during concomitant thiazolidinedione and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Thrombin Inhibitors: (Moderate) An additive risk of bleeding may be seen in patients receiving salicylates (e.g., aspirin) in combination with thrombin inhibitors. In addition, large doses of salicylates (3 to 4 g/day or more) can cause hypoprothrombinemia, an additional risk factor for bleeding. Nonsteroidal antiinflammatory drugs (NSAIDs) may

also increase bleeding risk when given with argatroban because of their potential to cause GI bleeding or inhibit platelet aggregation.
Thrombolytic Agents: (Moderate) Monitor for bleeding if the concomitant use of thrombolytics and aspirin is necessary. For patients who received a thrombolytic for acute ischemic stroke, avoid IV aspirin in the first 90 minutes following thrombolytic administration and consider delaying the initiation of aspirin therapy for at least 24 hours following thrombolytic administration. Both medications interfere with hemostasis and concomitant use increases the risk for bleeding.
Ticagrelor: (Moderate) Monitor for increased bleeding and avoid aspirin doses greater than 100 mg during ticagrelor coadministration. Lower aspirin doses (100 mg or less) are associated with a lower risk of bleeding. Additionally, maintenance doses of aspirin above 300 mg decreased ticagrelor efficacy compared with doses of 100 mg or less in a clinical trial.
Timolol: (Moderate) Concurrent use of beta-blockers with aspirin and other salicylates may result in loss of antihypertensive activity due to inhibition of renal prostaglandins and thus, salt and water retention and decreased renal blood flow.
Tipranavir: (Moderate) Caution should be used when administering tipranavir to patients receiving platelet inhibitors. In clinical trials, there have been reports of intracranial bleeding, including fatalities, in HIV infected patients receiving tipranavir as part of combination antiretroviral therapy. In many of these reports, the patients had other medical conditions (CNS lesions, head trauma, recent neurosurgery, coagulopathy, hypertension, or alcoholism/alcohol abuse) or were receiving concomitant medications, including platelet inhibitors, that may have caused or contributed to these events.
Tirofiban: (Moderate) Unless contraindicated, aspirin is used in combination with tirofiban. However, both drugs are associated with bleeding. Monitor for bleeding during concomitant therapy.
Tirzepatide: (Moderate) Monitor blood glucose during concomitant incretin mimetic and salicylate use. Concomitant use may cause an increased blood glucose-lowering effect with risk of hypoglycemia.
Tobramycin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents like the aminoglycosides may lead to additive nephrotoxicity.
Tolmetin: (Major) The concurrent use of aspirin with other NSAIDs should be avoided because this may increase bleeding or lead to decreased renal function. The use of salicylates together with NSAIDs can also lead to additive GI toxicity.
Torsemide: (Moderate) Salicylates may decrease the diuretic, natriuretic, and antihypertensive actions of diuretics, possibly through inhibition of renal prostaglandin synthesis. Patients receiving loop diuretics and salicylates should be monitored for changes in the effectiveness of their diuretic therapy.
Trandolapril: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE

inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation.
Trandolapril; Verapamil: (Moderate) Aspirin, ASA may reduce the vasodilatory efficacy of ACE inhibitors by inhibiting the synthesis of vasodilatory prostaglandins. This interaction has been documented primarily in heart failure patients. However, the established benefits of using aspirin in combination with an ACE inhibitor in patients with ischemic heart disease and left ventricular dysfunction generally outweigh this concern. Patients receiving concurrent salicylates and ACE inhibitor therapy should be monitored for antihypertensive or vasodilatory efficacy; the dose of the ACE inhibitor can be adjusted if indicated based on clinical evaluation. (Minor) In a few reported cases, coadministration of verapamil with aspirin, ASA has led to increased bleeding times greater than observed with aspirin alone. The exact mechanism and clinical significance of this interaction is unknown.
Trastuzumab; Hyaluronidase: (Minor) Salicylates, when given in large systemic doses, may render tissues partially resistant to the action of hyaluronidase. Patients receiving these medications may require larger amounts of hyaluronidase for equivalent dispersing effect.
traZODone: (Moderate) Monitor for signs and symptoms of bleeding during concomitant trazodone and salicylate use due to increased risk for bleeding. Serotonin release by platelets plays an important role in hemostasis. Epidemiological studies have demonstrated an association between use of psychotropic drugs that interfere with serotonin reuptake and the occurrence of upper gastrointestinal bleeding. Altered anticoagulant effects, including increased bleeding, have been reported when serotonin norepinephrine reuptake inhibitors are coadministered with another anticoagulant.
Triamcinolone: (Moderate) Monitor for gastrointestinal toxicity during concurrent corticosteroid and salicylate use. Concomitant use increases the risk of GI bleeding. In patients receiving concomitant corticosteroids and chronic use of salicylates, withdrawal of corticosteroids may result in salicylism because corticosteroids enhance renal clearance of salicylates and their withdrawal is followed by return to normal rates of renal clearance.
Triamterene: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant triamterene and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition

of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Triamterene; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention. (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant triamterene and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Tromethamine: (Moderate) Concurrent administration of high doses of alkalinizing agents may increase urine pH and decrease serum salicylate levels by decreasing renal tubular reabsorption of salicylic acid.
Valproic Acid, Divalproex Sodium: (Moderate) Concurrent salicylate therapy can increase the free-fraction of valproic acid, causing possible valproic acid toxicity. Valproic acid levels should be monitored when these agents are used concomitantly.
Valsartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor blood pressure as well as for signs of worsening renal function and loss of diuretic efficacy, including antihypertensive effects, during concomitant thiazide diuretic and salicylate use. Salicylate use decreases glomerular filtration rate and renal blood flow, and concomitant diuretic use may increase the risk of this reaction. Salicylates may diminish the effectiveness of diuretics due to inhibition of renal prostaglandins, leading to decreased renal blood flow and salt and fluid retention.
Vancomycin: (Minor) Due to the inhibition of renal prostaglandins by salicylates, concurrent use of salicylates and other nephrotoxic agents, such as vancomycin, may lead to additive nephrotoxicity.
Varicella-Zoster Virus Vaccine, Live: (Major) No adverse events associated with the use of salicylates after varicella vaccination have been reported. However, the manufacturer of varicella virus vaccine live recommends the avoidance of salicylates or aspirin, ASA use for 6 weeks after vaccination. Reye's syndrome, which exclusively affects children under 15 years old, has been associated with aspirin use following active varicella infection. Vaccination with close clinical monitoring is recommended for children who require therapeutic aspirin, ASA therapy; according to the CDC the use of attenuated, live varicella virus vaccine is thought to present less risk than natural varicella disease to such children.

Venlafaxine: (Moderate) Platelet aggregation may be impaired by serotonin norepinephrine reuptake inhibitors (SNRIs) due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin, ASA or other salicylates which affect hemostasis. Patients should be instructed to monitor for signs and symptoms of bleeding while taking an SNRI with medications which impair platelet function and to promptly report any bleeding events to the practitioner.
Verapamil: (Minor) In a few reported cases, coadministration of verapamil with aspirin, ASA has led to increased bleeding times greater than observed with aspirin alone. The exact mechanism and clinical significance of this interaction is unknown.
Verteporfin: (Moderate) Use caution if coadministration of verteporfin with aspirin is necessary due to the risk of decreased verteporfin efficacy. Verteporfin is a light-activated drug. Once activated, local damage to neovascular endothelium results in a release of procoagulant and vasoactive factors resulting in platelet aggregation, fibrin clot formation, and vasoconstriction. Concomitant use of drugs that decrease platelet aggregation like aspirin could decrease the efficacy of verteporfin therapy.
Vilazodone: (Moderate) Patients should be instructed to monitor for signs and symptoms of bleeding while taking vilazodone concurrently with salicylates or other platelet inhibitors and to promptly report any bleeding events to the practitioner. Platelet aggregation may be impaired by vilazodone due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving platelet inhibitors (e.g., aspirin, cilostazol, clopidogrel, dipyridamole, ticlopidine, platelet glycoprotein IIb/IIIa inhibitors).
vinCRIStine Liposomal: (Moderate) Acidification of the urine may increase serum concentrations of salicylates by increasing tubular reabsorption of salicylates, however, this interaction is not likely to be clinically significant since the urine is normally acidic.
Vonoprazan; Amoxicillin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Vonoprazan; Amoxicillin; Clarithromycin: (Minor) Due to the high protein binding of aspirin, it could displace or be displaced from binding sites by other highly protein-bound drugs, such as penicillins. Also, aspirin may compete with penicillin for renal tubular secretion, increasing penicillin serum concentrations. Overall, this combination should be used with caution and patients monitored for increased side effects.
Vorapaxar: (Moderate) Although indicated for concomitant use, both vorapaxar and aspirin are associated with bleeding. Monitor for bleeding during concomitant therapy.

Vortioxetine: (Moderate) Platelet aggregation may be impaired by vortioxetine due to platelet serotonin depletion, possibly increasing the risk of a bleeding complication (e.g., gastrointestinal bleeding, ecchymoses, epistaxis, hematomas, petechiae, hemorrhage) in patients receiving aspirin or other salicylates. Bleeding events related to drugs that inhibit serotonin reuptake have ranged from ecchymosis to life-threatening hemorrhages. Patients should be instructed to monitor for signs and symptoms of bleeding while taking vortioxetine concurrently with aspirin products and to promptly report any bleeding events to the practitioner.
Warfarin: (Major) Co-administration of aspirin and warfarin is associated with an increased risk of bleeding. Consider alternate therapy for aspirin for analgesic or antipyretic uses. If aspirin and warfarin are coadministered, monitor the patient for signs or symptoms of bleeding. Gastrointestinal irritation and impaired hemostasis secondary to platelet inhibition have been observed with relatively small doses of aspirin. In addition, aspirin may displace warfarin from protein binding sites leading to increased anticoagulation effects. Large doses (more than 3 to 4 g/day) of aspirin can cause hypoprothrombinemia, an additional risk factor for bleeding; hypoprothrombinemia has also been reported with aspirin doses less than 2 g/day. Lower doses (less than 100 mg) of aspirin are recommended for use in combination with aspirin for the prevention of cardiovascular events in specific cases, including in patients with mechanical mitral or aortic valve or atrial fibrillation after percutaneous coronary intervention or revascularization. The addition of warfarin to aspirin and a P2Y12 inhibitor in patients after ST-elevation myocardial infarction should be limited to situations where the risk of systemic or venous thromboembolism or stent thrombosis is considered to exceed that of bleeding. Data regarding the benefit vs. risk of combination therapy for other cardiovascular conditions remains unclear.
Zafirlukast: (Minor) Coadministration of aspirin may increase plasma concentrations of zafirlukast. The potential clinical sequelae of increased zafirlukast concentrations are not known.
Zoledronic Acid: (Moderate) Monitor renal function during concomitant zoledronic acid and aspirin use due to risk for additive nephrotoxicity.

Adverse Reaction

agitation, cerebral edema, coma, confusion, dizziness, headache, hearing loss, lethargy, seizures, tinnitus

Central nervous system adverse effects reported with aspirin include agitation, cerebral edema, coma, confusion, dizziness, headache, lethargy, and seizures. Tinnitus and hearing loss may occur in patients receiving high-dose and/or long-term aspirin therapy. Discontinue aspirin if tinnitus or hearing loss occurs. These effects are early

manifestations of salicylate toxicity. However, hearing loss has occurred in patients at low serum salicylate concentrations. Tinnitus and hearing loss are usually dose-related and reversible upon dose reduction or discontinuation. Tinnitus is commonly associated with salicylate concentrations more than 200 to 300 mcg/mL. Maximum hearing loss occurs most frequently at salicylate concentrations of 400 mcg/mL or more. The association of hearing loss and regular analgesic therapy (including aspirin) was prospectively assessed over 18 years in a study of 26,917 males age 40 to 74 years at study enrollment. During 369,079 person-years of follow-up, 3,488 cases of hearing loss were reported. After adjustment for confounders, the hazard ratio (HR) for aspirin associated hearing loss was 1.12 (95% CI 1.04 to 1.2, p = 0.005) in patients who were regular users of the drug (2 times weekly or more) compared to those with less use. Men who used aspirin regularly for 1 to 4 years were 28% (17% to 40%) more likely to develop hearing loss than those without regular use; the risk of hearing loss did not further increase with longer duration of use. Regular users of aspirin younger than 60 years were 33% more likely to experience hearing loss compared to non-regular users; no association occurred in those 60 years and older. This study does suggest association; however, data are based on patient reporting of the outcomes. Information regarding noise exposure and analgesic doses was not provided. Conversely, a similar prospective analysis conducted over 14 years in 62,261 women did not demonstrate an independent association of aspirin therapy with hearing loss. Salicylate ototoxicity may result from damage to the cochlea or auditory nerve through multiple mechanisms, including reduced cochlear blood flow, impairment of outer hair cell function, or inhibition of prostaglandin-forming cyclooxygenase.

abdominal pain, anorexia, dyspepsia, elevated hepatic enzymes, gastritis, GI bleeding, hepatitis, nausea, peptic ulcer, vomiting

Gastrointestinal adverse effects reported with aspirin include abdominal pain, anorexia, dyspepsia, elevated hepatic enzymes, gastritis, GI bleeding, hepatitis, nausea, and vomiting. Aspirin may cause peptic ulcer.

Reye's syndrome

Reye's syndrome has been reported with aspirin use. Aspirin may increase the risk of developing Reye syndrome, a rare but serious disease which can follow flu or chicken pox in children and adolescents. Reye syndrome has been reported in children of all ages; however, most of the reported cases have occurred in children 5 to 10 years. Data are not strong to support a dose-dependent association with Reye syndrome; however, a case-controlled study reported that patients who developed Reye syndrome (n = 27) had received larger doses for a longer duration compared with controls who did not develop Reye syndrome. Of the patients who developed Reye syndrome, 67% were

receiving more than 20 mg/kg/day of salicylates compared with only 22% of controls. Reye syndrome is a multisystem disorder evidenced by persistent vomiting, altered sensorium, elevated hepatic enzymes, hypoprothrombinemia, hyperammonemia, convulsions, and encephalopathy. If children are receiving chronic aspirin therapy, discontinue aspirin immediately if a fever develops, and do not resume aspirin until diagnosis confirms that the febrile viral illness has run its course and the absence of Reye syndrome.

interstitial nephritis, proteinuria, renal failure, renal papillary necrosis

Renal adverse effects reported with aspirin include interstitial nephritis, renal papillary necrosis, proteinuria, renal insufficiency and acute renal failure.

pruritus, rash

Dermatologic adverse reactions reported with aspirin include rash, pruritus, and purpura.

hyperuricemia

Salicylates, such as aspirin, have dose-dependent effects on plasma uric acid concentrations. At low doses (1 to 2 g/day) decreased urate excretion and hyperuricemia may be seen. Intermediate salicylate doses (2 to 3 g/day) usually do not alter urate excretion, and high doses of salicylates (more than 3 g/day) induce uricosuria and lower plasma uric acid concentrations.

intracranial bleeding

Intracranial bleeding has been reported with aspirin use.

dehydration, diaphoresis, hyperglycemia, hypernatremia, hyperventilation, hypoglycemia, hypokalemia, metabolic acidosis, respiratory alkalosis

At therapeutic doses, salicylates such as aspirin cause changes in acid/base balance and electrolytes resulting in respiratory alkalosis. In patients with normal renal and respiratory function, this is usually compensated for appropriately. Severe acid/base disturbances may occur during salicylate toxicity. Infants and children with salicylate toxicity rarely present clinically with respiratory alkalosis. As salicylate toxicity progresses, changes resembling metabolic acidosis are present (e.g., low blood pH, low plasma bicarbonate concentrations, and normal or nearly normal plasma PaCO2). In

reality, a combination of respiratory acidosis and metabolic acidosis is present. Alterations in water and electrolyte balance also occur in salicylate toxicity. Dehydration due to salicylate-induced diaphoresis and hyperventilation occurs. Since more water than electrolytes are lost, dehydration is associated with hypernatremia. Other laboratory changes noted in salicylate toxicity include hyperglycemia or hypoglycemia (especially in children), ketonuria, hypokalemia, and proteinuria. Prolonged exposure to high doses of salicylates also causes hypokalemia through both renal and nonrenal losses. Hyperventilation occurs due to direct stimulation of the respiratory center in the medulla. At high salicylate plasma concentrations (350 mcg/mL or more), marked hyperventilation will occur, and at serum concentrations of about 500 mcg/mL, hyperpnea will be seen. At high or prolonged doses, salicylates also have a depressant effect on the medulla. Toxic doses of salicylates cause central respiratory depression as well as cardiovascular collapse secondary to vasomotor depression. Since enhanced CO2 production continues, respiratory acidosis occurs.

medication overuse headache, withdrawal

Overuse of drugs for treating acute headaches, including aspirin, may lead to medication overuse headache. Patients may experience migraine-like daily headaches or a significant increase in migraine attack frequency. Discontinuation of the overused drug and treatment of withdrawal symptoms (e.g., transient worsening of headache) may be necessary. Advise patients about the risks of medication overuse (e.g., use of aspirin for at least 15 days/month or any combination of therapy for at least 10 days/month) and encourage them to keep a written record of headache frequency and drug use.

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), a multi-organ hypersensitivity reaction, has occurred with NSAIDs. Some of these events have been life-threatening or fatal. DRESS typically presents as fever, rash, and/or lymphadenopathy in conjunction with other organ system involvement including hepatitis, nephritis, hematologic abnormalities, myocarditis, or myositis sometimes resembling an acute viral infection. Eosinophilia is often present. Early manifestations such as fever and lymphadenopathy may be present without evidence of a rash. Discontinue the NSAID in patients presenting with such signs and symptoms in whom an alternative etiology cannot be identified.

infertility

NSAIDs may delay or prevent prostaglandin-mediated rupture of ovarian follicles, which has been associated with reversible infertility. Small studies with NSAIDs demonstrated

a reversible delay in ovulation. Consider withdrawal of NSAIDs in people who may become pregnant who have difficulties conceiving or who are undergoing infertility evaluation.

Description

Aspirin is an oral and rectal nonsteroidal antiinflammatory drug (NSAID) indicated for the temporary relief of minor aches and pains associated with headache, backache, muscular aches, a cold, toothache, minor pain of arthritis, premenstrual and menstrual cramps, to reduce the risk of death and myocardial infarction (MI) in patients with chronic coronary artery disease, such as patients with a history of MI or angina pectoris or chronic stable angina, and to reduce the risk of death and recurrent stroke in patients who have had an ischemic stroke or transient ischemic attack (TIA). Aspirin increases the risk of bleeding and may cause gastric ulceration and bleeding. Reye's syndrome, a potentially fatal disease, has been associated with aspirin use after active varicella infection or other viral illnesses in children; hence, aspirin use is children is primarily limited to the treatment of Kawasaki disease, for thrombosis prophylaxis, particularly in those with congenital heart disease after cardiac surgery, and for the treatment and secondary prevention of arterial ischemic stroke. Guidelines for the treatment of juvenile idiopathic arthritis in children no longer recommend aspirin as a treatment option due to the availability of other NSAIDs (i.e., ibuprofen, naproxen) that are just as effective, safer, and better tolerated. Observational studies have suggested that aspirin reduces the risk of colorectal cancer. However, long-term follow-up of the randomized Physicians' Health Study found no association between aspirin use and colorectal cancer. In contrast, randomized trials have shown that aspirin reduces the risk of recurrent adenomas in persons with a history of colorectal cancer or adenomas. The role of aspirin in the chemoprevention of colorectal cancer, either as primary or secondary prophylaxis, has not been determined.

Mechanism Of Action

The activity of aspirin is due to its ability to inhibit cyclooxygenase (COX). Cyclooxygenase is responsible for the conversion of arachidonic acid to prostaglandin G2 (PGG-2), the first step in prostaglandin synthesis and precursor to prostaglandins of the E and F series. Cyclooxygenase exists in 2 isozymes: cyclooxygenase-1 (COX-1) and cyclooxygenase-2 (COX-2). In vivo, aspirin is hydrolyzed to salicylic acid and acetate. However, hydrolysis is not required for aspirin activity. Aspirin irreversibly inhibits COX by acetylation of a specific serine moiety (serine 530 of COX-1 and serine 516 of COX-2). Aspirin is about 170-times more potent in inhibiting COX-1 than COX-2. In comparison,

salicylic acid has little or no ability to inhibit COX in vitro despite inhibiting prostaglandin synthesis at the site of inflammation in vivo. The exact mechanism of prostaglandin inhibition by salicylic acid is unclear; however, salicylates produce the majority of classic NSAID effects. Theories regarding the potential mechanism for salicylic acid include inactivation of transcriptional regulatory proteins (e.g., NF-kappaB), which regulate expression of inflammatory proteins. Aspirin appears to inhibit COX through two pathways and seems to have a different mechanism of action than other salicylates. Aspirin does not inhibit the peroxidase activity of COX and does not suppress leukotriene synthesis by lipoxygenase pathways.

Antithrombotic Actions: Aspirin-induced inhibition of thromboxane A2 (TXA2) and prostacyclin (PGI-2) has opposing effects on hemostasis. TXA2 is a potent vasoconstrictor and platelet agonist, while PGI-2 inhibits platelet aggregation and vascular smooth muscle contraction. However, data suggest that the effects of aspirin-induced TXA2 inhibition predominate clinically. This may be due to the ability of vascular endothelial cells to regenerate new COX and recover normal function, while COX inhibition in platelets is irreversible due to the limited amount of mRNA and protein synthesis in these cells. This distinction also allows for the use of very low doses of aspirin to retard platelet aggregation. The antithrombotic actions of aspirin are primarily mediated by COX-1 inhibition; COX-1 produces TXA2. Aspirin may also inhibit platelet activation by neutrophils. The antiplatelet effects of aspirin result in a prolonged bleeding time, which returns to normal roughly 36 hours after the last dose of the drug. Antiplatelet effects occur before acetylsalicylic acid is detectable in the peripheral blood due to exposure of platelets in the portal circulation. In very high and toxic doses, aspirin also exerts a direct inhibitory effect on vitamin K-dependent hemostasis by inhibiting the synthesis of vitamin K-dependent clotting factors. Prothrombin synthesis is impaired, resulting in hypoprothrombinemia.

Anti-inflammatory Actions: The antiinflammatory action of aspirin is believed to be a result of peripheral inhibition of COX-1 and COX-2, but aspirin may also inhibit the action and synthesis of other mediators of inflammation. It is thought that COX-2 is the more important pathway for the inflammatory response since COX-2 is inducible in settings of inflammation by cytokines. Inhibition of COX-2 by aspirin suppresses the production of prostaglandins of the E and F series. These prostaglandins induce vasodilation and increase tissue permeability, which, in turn, promotes the influx of fluids and leukocytes. Ultimately, the classic symptoms of inflammation result: swelling, redness, warmth, and pain. Aspirin does not only decrease capillary permeability (which reduces swelling and the influx of inflammatory mediators), but it can also reduce the release of destructive enzymes from lysozymes.

Analgesic Actions: Salicylates are effective in cases where inflammation has caused sensitivity of pain receptors (hyperalgesia). It appears prostaglandins, specifically prostaglandins E and F, are responsible for sensitizing the pain receptors; therefore,

salicylates have an indirect analgesic effect by inhibiting the production of further prostaglandins and do not directly affect hyperalgesia or the pain threshold. Salicylates may also interfere with pain perception centrally by activity within the hypothalamus. The total serum salicylate concentrations associated with analgesic activity are 30 to 100 mcg/mL.

Antipyretic Actions: Salicylates promote a return to a normal body temperature set point in the hypothalamus by suppressing the synthesis of prostaglandins, specifically PGE-2, in circumventricular organs in and near the hypothalamus. Salicylates rarely decrease body temperature in afebrile patients. Paradoxically, toxic doses of salicylates may increase body temperature by increasing oxygen consumption and metabolic rate. The total serum salicylate concentrations associated with antipyretic activity are 30 to 100 mcg/mL.

Gastrointestinal Effects: Adverse gastrointestinal effects from salicylates may be mediated through decreased prostaglandin synthesis due to inhibition of COX-1. A direct irritant effect on gastric mucosa may also be involved. Salicylates increase the permeability of the gastric mucosa to cations, thus increasing the entry of acid into the mucosa. Salicylates are also known to stimulate the chemoreceptor trigger zone, resulting in nausea and vomiting.

Respiratory Effects: The respiratory effects of salicylates lead to acid/base changes and alterations in electrolyte and water balance. Salicylates stimulate respiration directly and indirectly resulting in respiratory alkalosis. This is caused by a salicylate-induced increase in oxygen consumption, primarily in skeletal muscle, leading to increased carbon dioxide production and respiratory stimulation. Increased alveolar ventilation balances the increased carbon dioxide production; therefore, plasma carbon dioxide ($PaCO_2$) does not change. Salicylate-induced respiratory alkalosis is compensated for by increasing renal excretion of bicarbonate, which is accompanied by increased sodium and potassium excretion. The serum bicarbonate concentration is then lowered and the serum pH returns to normal (i.e., compensated respiratory alkalosis). However, if the respiratory response to hypercapnia has been depressed (e.g., administration of a barbiturate or opiate agonist), salicylates will cause a significant increase in $PaCO_2$ and respiratory acidosis. Hyperventilation also occurs due to direct stimulation of the respiratory center in the medulla. At high salicylate plasma concentrations (350 mcg/mL or more), marked hyperventilation will occur, and at serum concentrations of about 500 mcg/mL, hyperpnea will be seen. Finally, at high-therapeutic and at toxic doses, aspirin can affect oxidative phosphorylation, however, this action is insignificant at lower doses. Other changes in acid-base status (e.g., metabolic and respiratory acidosis) and electrolyte and water balance (hypokalemia, hypernatremia, dehydration) may be seen during salicylate intoxication.

Renal Effects: In addition to changes in sodium and fluid status secondary to acid/base changes, salicylates may decrease renal blood flow and glomerular filtration rate, which

may be accompanied by water and potassium retention, in sodium-restricted patients and patients with impaired renal function or hypovolemic states. Changes in renal function are due to inhibition of renal prostaglandin synthesis, which increase renal blood flow and maintain normal renal function. Salicylate-induced renal effects are uncommon in patients with normal renal function.
Uricosuric Effects: Salicylates act on the renal tubules to affect uric acid excretion. Lower doses (e.g., 1 to 2 g/day) of salicylates inhibit the active secretion of uric acid into the urine via the proximal tubules. However, high doses (more than 3 g/day) of salicylates inhibit the tubular reabsorption of uric acid, resulting in a uricosuric effect. Uric acid secretion is not changed at intermediate dosages. While once used for their uricosuric properties, other agents have replaced salicylates for this purpose.

Pharmacokinetics

Aspirin is administered orally or rectally. Salicylic acid is widely distributed with high concentrations in the liver and kidney. During chronic administration, salicylate concentrations in the fetus may be higher than those in the mother. Aspirin is poorly protein-bound as compared to salicylic acid. However, aspirin may acetylate albumin, resulting in changes the ability of albumin to bind other drugs. Protein binding of salicylic acid to albumin varies with serum salicylate and albumin concentrations. At salicylate concentrations of 100 mcg/mL or less, salicylic acid is 90% to 95% protein-bound; approximately 70% to 85% protein-bound at 100 to 400 mcg/mL; and only 20% to 60% protein-bound at serum concentrations of more than 400 mcg/mL. Patients with low serum albumin have higher free salicylate concentrations.

Aspirin has a half-life of 15 to 20 minutes in adults as it is rapidly hydrolyzed by the liver to salicylic acid. Salicylic acid is primarily metabolized in the liver. Metabolites include salicyluric acid (glycine conjugate), the ether or phenolic glucuronide, and the ester or acyl glucuronide. In addition, a small amount is metabolized to gentisic acid (2,5-dihydroxybenzoic acid) and 2,3-dihydroxybenzoic and 2,3,5-dihydroxybenzoic acids. Salicyluric acid and salicyl phenolic glucuronide are formed via saturable enzyme pathways, and therefore, exhibit non-linear pharmacokinetics. The elimination half-life of salicylic acid varies with dosage. After a single low dose, the serum half-life of salicylic acid is 2 to 3 hours, but can increase to 12 hours with anti-inflammatory doses and up to 15 to 30 hours after overdoses. Because of decreased serum protein binding, the effect of increasing doses is more pronounced on free salicylate concentrations than total salicylate concentrations. Approximately 80% to 100% of the salicylic acid from a single salicylate dose is excreted within 24 to 72 hours in the urine as free salicylic acid (10%), salicyluric acid (75%), salicylic phenolic (10%) and acyl (5%) glucuronides, and gentisic acid (less than 1%). The excretion of free salicylic acid is variable and depends upon the dose and the urinary pH. In alkaline urine, more than 30% of the dose may be

eliminated as free salicylic acid, but in acidic urine only about 2% is eliminated as free salicylic acid.

Affected cytochrome P450 isoenzymes and drug transporters: none

Route-Specific Pharmacokinetics

- **Oral Route**

Aspirin is absorbed orally via passive diffusion as unchanged drug and as hydrolyzed salicylic acid from the upper intestine and partly from the stomach. Approximately 70% of an aspirin dose reaches the circulation unchanged; the remaining 30% is hydrolyzed to salicylic acid during absorption by esterases in the GI tract, plasma, or liver. The rate of absorption is dependent upon many factors including oral formulation, gastric and intestinal pH, gastric emptying time, and the presence of food. Aspirin is rapidly absorbed after oral administration, and bioavailability of regular aspirin in adults is approximately 40% to 50%. Effervescent and soluble tablets are most rapidly absorbed, followed by uncoated or film-coated tablets, and then enteric-coated tablets and extended-release formulations. The absorption from enteric-coated tablets and sustained-release preparations is delayed and bioavailability is significantly lower compared with regular aspirin. Peak plasma salicylate concentrations occur in approximately 30 to 60 minutes for effervescent tablets, 45 to 120 minutes for film-coated tablets, 4 to 12 hours for extended-release tablets, and 8 to 14 hours for enteric-coated tablets. Food decreases the rate, but not the extent, of absorption. Salicylic acid is more ionized as the pH increases; however, a rise in pH increases the solubility of ionized salicylic acid and increases the dissolution of aspirin tablets. The overall effect of increased pH is an increase in absorption. Time to peak aspirin concentrations is 15 to 240 minutes depending upon the formulation. Plasma aspirin concentrations decrease as salicylic acid concentrations increase. Steady-state salicylate serum concentrations are similar after administration of plain, uncoated tablets and enteric-coated tablets.

- **Other Route(s)**

Rectal Route

The bioavailability of aspirin after rectal administration in adults has been reported to be 20% to 40%. Peak concentrations are reached approximately 4 hours after rectal administration in adults. Limited pharmacokinetic data in 8 children (5 to 9 years) revealed that the absorption of aspirin was very slow after rectal administration and was highly dependent on retention time. In children that retained the suppository for 5 hours or less, urinary recovery was 54% to 64%. Therefore, aspirin given rectally may not attain effective serum concentrations.

- **Hepatic Impairment**

Pharmacokinetic data are unavailable in patients with hepatic impairment; however, aspirin is extensively metabolized in the liver and patients with hepatic impairment may have decreased elimination.

- **Renal Impairment**

Pharmacokinetic data are unavailable in patients with renal impairment. Aspirin is renally excreted and patients with renal impairment may have decreased elimination. Aspirin is 50% to 100% hemodialyzable.

- **Pediatrics**

Neonates would be expected to have a slower clearance of aspirin due to their immature hepatic function. Pharmacokinetic data are very limited in children. Data from 10 children (2 to 7 years) who received aspirin revealed a mean elimination half-life for salicylic acid of 3.4 hours. This is similar to what has been reported in adults.

The pharmacokinetics of aspirin are altered in children with Kawasaki disease. These patients have been shown to achieve lower salicylate concentrations compared with healthy children receiving the same aspirin dose due to a combination of impaired bioavailability and/or increased clearance.

Administration

For storage information, see the specific product information within the How Supplied section.

Oral Administration

Administer with food or large amounts (240 mL) of water or milk to minimize gastric irritation.

Oral Solid Formulations

Film-coated tablets: May help to reduce the unpleasant taste or aftertaste, burning in the throat, or difficulty in swallowing associated with uncoated tablets.
Enteric-coated or extended-release tablets: Swallow whole; do not crush, cut, or chew. May help to reduce gastric irritation and/or symptomatic GI disturbances associated with uncoated tablets.
Chewable tablets: May be chewed, crushed, and/or dissolved in a liquid, or swallowed whole, followed by approximately 120 mL of water, milk, or fruit juice immediately after administration.
Capsules: Swallow whole; do not crush, cut, or chew.

Rectal Administration

For use in patients unable to take or retain oral aspirin; however, absorption may be slow and incomplete. Do not use aspirin tablets rectally because they are likely to cause irritation and erosion of rectal mucosa.
Instruct patient or caregiver on proper use of suppository.
Prior to insertion, carefully remove the wrapper. Avoid excessive handling as to avoid melting of the suppository.
If suppository is too soft to insert, chill in the refrigerator for 30 minutes or run cold water over it before removing the wrapper.
Moisten the suppository with cool water prior to insertion.
Have patient lie down on their side, usually in the Sim's lateral position to provide support and comfort.
Apply gentle pressure to insert the suppository completely into the rectum, pointed end first, using a gloved, lubricated index finger.
After insertion, keep the patient lying down to aid retention and gently hold the buttock cheeks close to keep the child from immediately expelling the suppository. The suppository must be retained in rectum to ensure complete absorption.

Maximum Dosage Limits

Aspirin dosage must be individualized and is highly variable depending on the indications, coexisting conditions, and on patient response.

Dosage Forms

- Alka-Seltzer Effervescent Tablet
- Alka-Seltzer Effervescent Tablet
- Alka-Seltzer Effervescent Tablet (Lemon Lime)
- Alka-Seltzer Plus Cold and Cough Effervescent Tablet (Citrus)
- Alka-Seltzer Plus Cold Day Night Effervescent Tablet
- Alka-Seltzer Plus Cold Effervescent Tablet
- Alka-Seltzer Plus Cold Effervescent Tablet (Orange Zest)
- Alka-Seltzer Plus Cold Night Effervescent Tablet (Lemon)
- Anacin 400mg-32mg Tablet
- Aspir-Trin 650mg Tablet
- Aspirin 25mg, Dipyridamole 200mg Oral capsule, extended release
- Aspirin 300mg Rectal suppository
- Aspirin 325mg Oral tablet
- Aspirin 325mg Oral tablet, gastro-resistant

- Aspirin 385mg, Caffeine 30mg, Orphenadrine Citrate 25mg Oral tablet
- Aspirin 400mg, Caffeine 32mg Oral tablet
- Aspirin 500mg Oral tablet
- Aspirin 500mg Oral tablet, gastro-resistant
- Aspirin 650mg Oral tablet
- Aspirin 81mg Chewable tablet
- Aspirin 81mg Enteric Coated Tablet
- Aspirin 81mg Oral tablet, gastro-resistant
- Aspirin 81mg, Omeprazole 40mg Oral tablet, biphasic release
- Aspirin Bulk powder
- Bayer Advanced Aspirin 325mg Tablet
- Bayer Aspirin 325mg Tablet
- Bayer Aspirin 81mg Low Dose Chewable Tablet (Orange)
- Bayer Aspirin Plus 500mg Extra Strength Caplet
- Bayer Aspirin Regimen 325mg Caplet
- Bayer Aspirin Regimen 81mg Low Dose Chewable Tablet (Orange)
- Bayer Back and Body Pain 500mg-32.5mg Extra Strength Caplet
- Bayer Children's Aspirin 81mg Low Dose Chewable Tablet (Cherry)
- Bayer Extra Strength 500mg Caplet
- Bayer Extra Strength Plus 500mg Caplet
- Bayer Genuine Aspirin 325mg Tablet
- Bayer Genuine Aspirin 325mg Tablet
- Bayer Low Dose Aspirin Regimen 81mg Tablet
- Bayer Low Dose Aspirin Regimen 81mg Tablet
- Bayer Low Dose Aspirin Regimen 81mg Tablet
- Bayer Women's Aspirin 81mg Plus Calcium Buffered Caplet
- BC Arthritis Formula 1000mg-65mg Powder
- BC Headache 845mg-65mg Powder
- BC Headache 845mg-65mg Powder
- BC Headache Pain Relief 845mg-65mg Powder (Cherry)
- BC Headache Pain Relief 845mg-65mg Powder (Cherry)
- BC Headaches & Body Aches Fast Pain Relief 845mg-65mg Powder
- BC Headaches & Body Aches Fast Pain Relief 845mg-65mg Powder (Cherry)
- BC On-The-Go Arthritis Pain Fast Pain Relief 1000mg-65mg Powder
- BC On-The-Go Headaches & Body Aches Fast Pain Relief 845mg-65mg Powder
- BeneHealth Aspirin 81mg Low Dose Enteric Coated Tablet
- Bufferin 325mg Tablet
- Bufferin Extra Strength 500mg Tablet
- CAREALL Aspirin 81mg Chewable Tablet
- CVS Antacid & Pain Reliever Effervescent Tablet

- CVS Aspirin 325mg Buffered Tablet
- CVS Aspirin 325mg Coated Tablet
- CVS Aspirin 325mg Enteric Coated Tablet
- CVS Aspirin 325mg Tablet
- CVS Aspirin Adult Low Dose 81mg Enteric Coated Tablet
- CVS Aspirin Adult Low Strength 81mg Enteric Coated Tablet
- CVS Aspirin Low Dose 81mg Chewable Tablet (Cherry)
- CVS Aspirin Low Dose 81mg Chewable Tablet (Orange)
- CVS Aspirin Low Dose 81mg Enteric Coated Tablet
- CVS Aspirin Low Dose Sugar Free 81mg Enteric Coated Tablet
- CVS Aspirin Low Strength 81mg Chewable Tablet
- CVS Aspirin Low Strength 81mg Enteric Coated Tablet
- CVS Back and Body Pain 500mg-32.5mg Caplet
- CVS Cold Relief Effervescent Tablet
- CVS Enteric Aspirin 325mg Tablet
- CVS Fast Relief Antacid & Pain Reliever Effervescent Tablet
- CVS Fast Relief Antacid & Pain Reliever Extra Strength Effervescent Tablet
- CVS Genuine Aspirin 325mg Tablet
- DURLAZA 162.5mg Extended-Release Capsule
- Ecotrin 325mg Enteric Coated Tablet
- Ecotrin 325mg Enteric Coated Tablet
- Ecotrin 81mg Low Strength Enteric Coated Tablet
- Ecotrin 81mg Low Strength Enteric Coated Tablet
- Equaline Aspirin 81mg Chewable Tablet (Cherry)
- Equaline Aspirin 81mg Chewable Tablet (Orange)
- Equate Aspirin 325mg Safety Coated Tablet
- Equate Aspirin 325mg Safety-coated Tablet
- Equate Aspirin 81mg Enteric Coated Tablet
- Equate Aspirin Adult Low Dose 81mg Safety Coated Tablet
- Equate Aspirin Low Dose 81mg Chewable Tablet Triple Pack (Cherry)
- Equate Aspirin Low Dose 81mg Safety Coated Tablet
- Equate Aspirin Pain Reliever/Fever Reducer 325mg Coated Tablet
- Equate Effervescent Antacid Pain Relief 325mg-1000mg-1916mg Tablet
- Equate Low Dose Aspirin 81mg Chewable Tablet (Orange)
- Foster & Thrive Aspirin 325mg Coated Tablet
- Foster & Thrive Low Dose Aspirin 81mg Enteric Coated Tablet
- Foster & Thrive Low Dose Aspirin 81mg Enteric-Coated Tablet
- Foster & Thrive Regular Strength Aspirin 325mg Enteric Coated Tablet
- GNP Aspirin 325mg Coated Tablet
- GNP Aspirin 325mg Coated Tablet

- GNP Aspirin 325mg Enteric Coated Tablet
- GNP Aspirin 325mg Enteric Coated Tablet
- GNP Aspirin 325mg Tablet
- GNP Aspirin 325mg Tablet
- GNP Aspirin 81mg Chewable Tablet (Cherry)
- GNP Aspirin 81mg Chewable Tablet (Orange)
- GNP Aspirin 81mg Enteric Coated Tablet
- GNP Aspirin 81mg Enteric Coated Tablet
- GNP Aspirin 81mg Enteric Coated Tablet
- GNP Aspirin Low Dose 81mg Safety Coated Tablet
- GoodSense Aspirin 325mg Tablet
- GoodSense Aspirin 325mg Tablet
- GoodSense Aspirin 81mg Chewable Tablet
- GoodSense Aspirin 81mg Chewable Tablet (Cherry)
- GoodSense Aspirin 81mg Chewable Tablet (Orange)
- GoodSense Aspirin 81mg Safety Coated Tablet
- GoodSense Aspirin/Caffeine 400mg-32mg Tablet
- Goody's Hangover Pain Relief 1,000mg-150mg Oral Powder (Mixed Fruit Boost)
- Goody's Hangover Pain Relief Powder (Mixed Fruit Boost)
- Goody's Plus Headache Pain + Mental Alertness Fast Relief 1,000mg-150mg Oral Powder (Strawberry Citrus)
- Health Star Aspirin Adult Low Dose 81mg Enteric Coated Tablet
- Kroger Regular Strength Aspirin 325mg Coated Tablet
- Leader Adult Aspirin Regimen 81mg Enteric Coated Tablet
- Leader Antacid & Pain Reliever Extra Strength Effervescent Tablet
- Leader Aspirin 325mg Enteric Coated Tablet
- Leader Aspirin 325mg Tablet
- Leader Aspirin 81mg Chewable Tablet (Orange)
- Leader Aspirin 81mg Chewable Tablet (Orange)
- Leader Aspirin Adult Low Strength 81mg Enteric Coated Tablet
- Leader Aspirin Low Dose 81mg Chewable Tablet (Cherry)
- Leader Aspirin Low Dose 81mg Chewable Tablet (Orange)
- Leader Aspirin Regimen 81mg Enteric Coated Tablet
- Leader Aspirin Regimen Low Dose 81mg Enteric Coated Tablet
- Leader Back and Body Pain 500mg-32.5mg Caplet
- Leader Low Dose Adult Aspirin Regimen 81mg Enteric Coated Tablet
- Medique Medi-Seltzer Tablet
- MiniPrin Low Dose 81mg Enteric Coated Tablet
- NobleAid Aspirin 81mg EC Tablet
- NORGESIC 25mg-385mg-30mg Tablet

- Premier Value Antacid & Pain Relief Effervescent Tablet
- Premier Value Aspirin 325mg Enteric Coated Tablet
- Premier Value Aspirin 325mg Enteric Coated Tablet
- Premier Value Aspirin 325mg Tablet
- Premier Value Aspirin 81mg Chewable Tablet (Cherry)
- Premier Value Aspirin 81mg Chewable Tablet (Orange)
- Premier Value Aspirin 81mg Enteric Coated Tablet
- Premier Value Children's Aspirin 81mg Chewable Tablet (Orange)
- Premier Value Low Dose Aspirin 81mg Enteric Coated Tablet
- Publix Aspirin 325mg Coated Tablet
- Publix Aspirin 81mg Low Dose Chewable Tablet (Orange)
- Publix Enteric Aspirin 325mg Tablet
- Publix Fast Relief Effervescent Tablet
- Quality Choice Antacid & Pain Relief 1000mg-500mg-1985mg Extra Strength Effervescent Tablet
- Quality Choice Antacid & Pain Relief 325mg-1,000mg-1,916mg EffervescentTablet
- Quality Choice Aspirin 325mg Coated Tablet
- Quality Choice Aspirin 325mg Coated Tablet
- Quality Choice Aspirin 81mg Enteric Coated Tablet
- Quality Choice Aspirin Low Dose 81mg Chewable Tablet
- Quality Choice Aspirin Low Dose 81mg Chewable Tablet (Orange)
- Quality Choice Aspirin Low Dose 81mg Enteric Coated Tablet
- Quality Choice Cold Relief Plus 325mg-2mg-7.8mg Effervescent Tablet
- Quality Choice Cold Relief Plus 325mg-2mg-7.8mg Effervescent Tablet (Orange)
- Quality Choice Enteric Aspirin 325mg Tablet
- Quality Choice Low Dose Aspirin 81mg Chewable Tablet (Cherry)
- Quality Choice Low Dose Aspirin 81mg Enteric-Coated Tablet
- Quality Choice Pain Relief 845mg-65mg Powder
- RITE AID Aspirin 325mg Coated Tablet
- RITE AID Aspirin 325mg Tri-Buffered Tablet
- RITE AID Aspirin Low Dose 81mg Chewable Tablet (Cherry)
- RITE AID Aspirin Low Dose 81mg Chewable Tablet (Orange)
- RITE AID Aspirin Low Dose 81mg Enteric Coated Tablet
- RITE AID Aspirin Low Strength 81mg Enteric Coated Tablet
- RITE AID Low Dose Aspirin 81mg Chewable Tablet
- RITE AID Pain Relief Aspirin 325mg Enteric Coated Tablet
- Select Brand Aspirin 325mg Enteric Coated Tablet
- Select Brand Aspirin 325mg Tablet
- Select Brand Aspirin 81mg Enteric Coated Tablet
- Select Brand Children's Aspirin 81mg Chewable Tablet

- St. Joseph Adult Aspirin 81mg Chewable Tablet
- St. Joseph Aspirin Adult Low Strength 81mg Safety Coated Tablet
- Stanback Headache 845mg-65mg Powder
- Sunmark Low Dose Aspirin 81mg Tablet
- Today's Health Aspirin 325mg Tablet
- Today's Health Aspirin 81mg Chewable Tablet
- Today's Health Aspirin Low Dose 81mg Enteric Coated Tablet
- Today's Health Enteric Aspirin 325mg Tablet
- Top Care Aspirin 325mg Enteric Coated Tablet
- Top Care Aspirin 81mg Chewable Tablet
- Top Care Aspirin 81mg Chewable Tablet (Cherry)
- Top Care Aspirin 81mg Enteric Coated Tablet
- TopCare Aspirin 325mg Tablet
- TopCare Aspirin Low Dose 81mg Enteric Coated Tablet
- VAZALORE 325mg Liquid Filled Capsule
- VAZALORE 81mg Low Dose Liquid Filled Capsule
- Walgreens Antacid + Pain Relief Extra Strength 1000mg-500mg-1985mg Effervescent Tablet
- Walgreens Aspirin 325mg Enteric Coated Caplet
- Walgreens Aspirin 325mg Enteric Coated Tablet
- Walgreens Aspirin 325mg Tablet
- Walgreens Aspirin 325mg Tri-Buffered Tablet
- Walgreens Aspirin 81 81mg Enteric Coated Tablet
- Walgreens Aspirin 81mg Enteric Coated Tablet
- Walgreens Aspirin Adult Low Dose 81mg Chewable Tablet (Orange)
- Walgreens Children's Aspirin 81mg Chewable Tablet
- Walgreens Cold Relief Plus 325mg-2mg-7.8mg Effervescent Tablet
- Walgreens Dye-Free Aspirin 81mg Enteric Coated Tablet
- Walgreens Low Dose Aspirin 81 81mg Enteric Coated Tablet
- Walgreens Low Dose Aspirin 81mg Enteric Coated Tablet
- Walgreens Low Dose Aspirin 81mg Enteric-Coated Tablet
- Walgreens Low Dose Aspirin 81mg Tablet
- Walgreens Pain Relief Effervescent Tablet

Dosage Adjustment Guidelines

Hepatic Impairment

Avoid aspirin in patients with severe hepatic insufficiency. Patients with any degree of hepatic disease are at increased risk of salicylate-induced adverse reactions.

Renal Impairment

CrCl less than 10 mL/minute: Avoid analgesic doses. Use of low-dose aspirin for primary and secondary prevention of atherosclerotic events in patients with cardiovascular disease is recommended.

Intermittent hemodialysis
Avoid analgesic doses. Use of low-dose aspirin for primary and secondary prevention of atherosclerotic events in patients with cardiovascular disease is recommended. If use is necessary, doses should be administered after hemodialysis; aspirin is 50% to 100% dialyzable.

Continuous ambulatory peritoneal dialysis (CAPD)
Avoid analgesic doses. Use of low-dose aspirin for primary and secondary prevention of atherosclerotic events in patients with cardiovascular disease is recommended.

Continuous renal replacement therapy (CRRT)
No dosage adjustment needed; monitor serum salicylate concentrations if possible.

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